University of Glasgow | School of Computing Science

Predicting Mental Fatigue Using Physiological Sensors

Satveer Sahaas Ganasan
2785448g

Abstract

Mental fatigue impairs cognitive performance and decision-making, yet existing traditional monitoring methods such as EEG and self-report questionnaires remain impractical for continuous real-world use. This study investigates whether heart rate variability (HRV) and wrist skin temperature can be used to detect mental fatigue using machine learning and deep learning. Using the FatigueSet dataset (12 participants, 36 sessions), 22 HRV features were extracted from raw RR intervals and 6 temperature features were computed using 180-second sliding windows. Statistical analysis indicated that 26 of 28 features differed significantly ($p < 0.05$) between baseline and fatigued states, though these p-values should be interpreted with caution given the correlated nature of overlapping windows. Feature ablation identified a reduced subset of 13 features that outperformed the full set, with temperature change from baseline emerging as the most important predictor. HRV alone achieved F1 = 0.376 while temperature alone achieved 0.570, suggesting that the multimodal combination (F1 = 0.778) may outperform either modality in isolation. Unsupervised analysis using PCA and k-means clustering suggested that fatigue states are not naturally separable without labels (52.8% cluster alignment), supporting the use of a supervised approach. Deep learning models applied directly to raw RR intervals (CNN, LSTM, CNN-LSTM) achieved a best F1 score of 0.502, substantially below feature-based approaches. An MLP using the same extracted features achieved F1 = 0.754 (mean over 5 seeds), suggesting that the hand-crafted feature pipeline is the primary driver of performance in this dataset. Autoencoder-based feature compression did not improve upon the original features, further supporting the effectiveness of domain-specific feature engineering for small physiological datasets. These findings suggest the feasibility of non-invasive mental fatigue detection using wearable physiological signals, though validation on larger cohorts is needed before generalisability can be confirmed.

1 Introduction

Mental fatigue is a psychobiological state caused by sustained cognitive effort, characterised by reduced alertness, impaired decision-making, and diminished executive function [1]. Unlike sleepiness or boredom, which stem from circadian or motivational factors, mental fatigue emerges specifically from prolonged high cognitive demand and is best alleviated through rest [27]. Mental fatigue is also distinct from physical fatigue, which results from muscular exertion rather than cognitive intensity, and arises from engagement with tasks involving sustained attention, working memory, and executive control [23].

The consequences of undetected mental fatigue are significant. In safety-critical industries such as transportation, healthcare, and aviation, mental fatigue is a leading contributor to human error and accidents [3]. Fatigue-related road traffic accidents account for a significant proportion of crashes globally [25], and the World Health Organisation has recognised occupational burnout (a condition closely related to chronic mental fatigue) as an occupational phenomenon [31]. Despite its prevalence, mental fatigue remains difficult to measure objectively. Electroencephalography (EEG) provides direct neural correlates but requires laboratory conditions and specialised equipment [30]. Self-report instruments such as the Visual Analogue Scale (VAS) are subjective and cannot capture rapid fluctuations [15]. These limitations highlight the need for objective, continuous, and non-invasive monitoring methods.

This study investigates whether heart rate variability (HRV) and wrist skin temperature, recorded from wearable sensors, can be used to detect mental fatigue using machine learning and deep learning. Using the FatigueSet dataset [13] (12 participants, 36 sessions), the primary contributions are:

(1) A feature extraction pipeline computing 22 HRV features from raw RR intervals and 6 skin temperature features using empirically validated 180-second sliding windows with a 60-second step size.
(2) Statistical evidence that 26 of 28 features differ significantly ($p < 0.05$) between baseline and fatigued states, though these p-values should be interpreted with caution given the correlated nature of overlapping windows, with feature ablation identifying a reduced subset of 13 features that outperformed the full set under LOSO cross-validation.
(3) Evidence suggesting that the multimodal combination of HRV and temperature (F1 = 0.778) may outperform either modality in isolation (HRV-only F1=0.376, temperature-only F1=0.570).
(4) A rigorous evaluation using Leave-One-Subject-Out (LOSO) cross-validation with comprehensive comparison across traditional ML, deep learning, and autoencoder-based approaches.

1.1 Research Questions

(1) Can a combination of HRV and wrist skin temperature features distinguish between baseline and mentally fatigued states using supervised machine learning under cross-subject evaluation?
(2) Does a multimodal approach outperform single-modality models, and which features contribute most to classification performance?
(3) How do traditional machine learning methods compare with deep learning approaches for mental fatigue detection on small physiological datasets?

2 Background

This section reviews relevant background literature on physiological markers of mental fatigue, wearable sensing technologies, and machine learning approaches for fatigue detection, identifying key research gaps that motivate the present study.

2.1 Heart Rate Variability

Among the physiological signals accessible through wearable devices, heart rate variability (HRV) has received the most attention as a potential marker of cognitive and affective states. HRV refers to the variation in time intervals between consecutive heartbeats, quantified from the RR intervals of the electrocardiogram (ECG)

or from photoplethysmography (PPG) signals [26]. Contrary to a common misconception, a healthy heart does not beat with metronomic regularity; rather, the inter-beat intervals fluctuate from beat to beat in response to autonomic nervous system (ANS) modulation.

The ANS comprises two branches with generally opposing effects on cardiac function. The sympathetic nervous system (SNS) increases heart rate and reduces variability in response to stress, physical exertion, or perceived threat. The parasympathetic nervous system (PNS), mediated primarily through the vagus nerve, decreases heart rate and increases variability, promoting rest, recovery, and homeostasis [26]. The balance between these two branches is reflected in the frequency-domain decomposition of the HRV signal. Low-frequency (LF) power (0.04-0.15 Hz) reflects a combination of sympathetic and parasympathetic influences, while high-frequency (HF) power (0.15-0.40 Hz) is widely accepted as a marker of parasympathetic (vagal) activity, as it corresponds to the respiratory sinus arrhythmia [16].

A systematic review by Penfold et al. reported that decreased HRV is consistently associated with increased fatigue across multiple medical populations [22]. Research has consistently shown associations between sustained cognitive load and suppressed parasympathetic activity, leading to reductions in HRV metrics. Specifically, mental stress has been associated with decreased RMSSD (root mean square of successive differences), decreased HF power, and increased LF/HF ratio, reflecting vagal withdrawal and a shift towards sympathetic dominance [10]. Taelman et al. further reported that pNN50 decreased significantly and mean RR intervals shortened during mental stressor tasks [29]. Time-domain measures such as SDNN (standard deviation of NN intervals) capture overall variability from all sources, while frequency-domain and geometric measures (such as the triangular index) provide complementary perspectives on autonomic regulation. However, HRV is sensitive to numerous confounding factors including posture, physical activity, respiratory rate, emotional state, caffeine intake, and circadian rhythms [19], which can limit its reliability as a standalone fatigue marker.

2.2 Skin Temperature

Skin temperature, while less extensively studied in the context of mental fatigue compared to HRV, has been associated with changes in alertness, vigilance, and thermoregulatory control. The core body temperature follows a circadian rhythm that is closely linked to the sleep-wake cycle, with higher distal (skin) temperatures associated with decreased alertness and increased sleep propensity [24]. The underlying mechanism involves the redistribution of blood flow between the body core and the periphery. When peripheral blood vessels dilate (vasodilation), heat dissipation increases, skin temperature rises, and core temperature decreases, promoting drowsiness.

During sustained cognitive effort, the sympathetic nervous system induces peripheral vasoconstriction, reducing blood flow to the extremities and causing measurable decreases in wrist and fingertip skin temperature [9]. This thermoregulatory response is distinct from the temperature changes caused by physical exercise, which involve a different pattern of vasodilation for heat dissipation. Molina et al. reported that peripheral temperature dynamics change during prolonged psychomotor vigilance tasks, with temperature variations correlating with performance decrements [17]. The slow temporal dynamics of skin temperature (changing over minutes rather than beat-to-beat) complement the faster dynamics of HRV, suggesting that a multimodal approach combining both signals may capture different aspects of the fatigue response that neither modality can capture alone.

2.3 Wearable Physiological Monitoring

The emergence of wearable sensing technologies has created new opportunities for unobtrusive physiological monitoring outside laboratory settings. Modern wearable devices, including wristbands, chest straps, and ear-worn sensors such as OmniBuds [18], can continuously record a range of physiological signals including heart rate, heart rate variability (HRV), skin temperature, electrodermal activity (EDA), and inertial measurements [13]. These devices operate passively, require no conscious effort from the user after initial setup, and can collect data continuously over extended periods, making them well-suited for real-world fatigue monitoring applications. When combined with machine learning, such continuous physiological streams enable the possibility of automated, real-time fatigue detection without requiring user intervention or subjective self-assessment.

2.4 Machine Learning for Fatigue Detection

Machine learning (ML) has increasingly become the method of choice for detecting mental fatigue from physiological and behavioural data. A variety of algorithms, from traditional classifiers to deep learning models, have been applied, each presenting different trade-offs.

Many early and some current fatigue detection studies employ classical ML models such as Support Vector Machines (SVM), Random Forests (RF), Decision Trees (DT), and k-Nearest Neighbours (KNN). For example, a recent study monitoring nurse fatigue used HRV data and trained a fine-Gaussian SVM classifier, achieving approximately 81.5% accuracy and an F1 score of approximately 81% [8]. Butkeviciute et al. demonstrated that Random Forest classification could distinguish fatigued from non-fatigued states using ECG-based features with high reported accuracy [2]. However, these studies relied on single physiological modalities, which are susceptible to confounding factors such as physical exertion and inter-individual variability, and neither explicitly distinguished between mental and physical fatigue. Guo et al. addressed the single-modality limitation by classifying flight fatigue into three levels using LightGBM with combined HRV and respiratory features, achieving 88.6% accuracy across 90 pilots [7].

Multimodal approaches have shown promise for improving robustness in cognitive state detection. One study demonstrated that fusion models combining EEG, heart rate, galvanic skin response, and eye tracking can substantially improve classification of cognitive workload [20]. Their study introduced a hybrid machine learning pipeline that integrates supervised and unsupervised methods with a novel feature augmentation strategy based on clustering and centroid optimization. Similarly, Hirachan et al. reported that multimodal sensor fusion significantly increases accuracy in detecting cognitive workload [11]. Such approaches help mitigate the confounds of individual physiological signals, making models more robust to noise from any single modality. However, these studies primarily address momentary workload estimation rather than mental fatigue as a cumulative decline in cognitive performance over time, and the added complexity of synchronising multiple sensor streams at different sampling rates presents practical engineering challenges.

Because obtaining reliable ground-truth labels for mental fatigue is difficult (self-reports are subjective and performance-based labels are often confounded), unsupervised and semi-supervised learning methods have gained increasing attention. These methods aim to discover latent physiological states, patterns, or clusters corresponding to fatigue levels without requiring explicit labels. A recent study applied clustering (i.e. k-means, hierarchical clustering, DBSCAN) to a comprehensive set of linear, spectral and nonlinear HRV features and successfully identified latent groups corresponding to different physiological states [5]. This suggests the feasibility of unsupervised physiological state recognition purely from HRV, which is relevant when fatigue labels are unavailable. However, the validity of such approaches depends critically on the quality of the feature engineering, as clustering methods are sensitive to irrelevant or redundant features.

2.5 Leave-One-Subject-Out Cross-Validation

A fundamental challenge in physiological machine learning is ensuring that models generalise to unseen individuals rather than simply memorising the physiological patterns of participants seen during training. This is particularly critical for fatigue detection, where absolute HRV values and skin temperature baselines vary considerably between individuals due to differences in fitness level, age, autonomic nervous system tone, and circadian phase [26]. Standard k-fold cross-validation, which randomly partitions windows across participants, violates this requirement by allowing the model to train and test on windows from the same individual, producing inflated performance estimates that do not reflect real-world deployment [21].

Leave-One-Subject-Out (LOSO) cross-validation addresses this by treating each participant as a held-out test fold in turn, training on the remaining $n-1$ participants and testing on the left-out participant. This procedure ensures complete separation between training and test individuals, providing an unbiased estimate of cross-subject generalisation. LOSO is widely considered the appropriate evaluation protocol for physiological and affective computing studies [6], and has been adopted in fatigue and stress detection literature as the standard for reporting generalisable results. However, LOSO evaluation also reveals the inherent difficulty of the problem: performance typically drops substantially compared to within-subject evaluations, reflecting the challenge of building models that are robust to inter-individual physiological variability.

2.6 Research Gaps

Despite these advances, several important gaps remain. First, the majority of existing studies rely on a single physiological modality. Hafiz et al. [8] and Butkeviciute et al. [2] both used only ECG-derived features for fatigue recognition. Single-sensor approaches are vulnerable to confounding factors such as posture [19], physical activity, and emotional state. Few studies have investigated whether combining HRV with skin temperature improves classification, despite evidence that temperature correlates with vigilance decrements [17]. The present study addresses this gap by evaluating a multimodal approach and quantifying the relative contribution of each modality.

Second, many studies fail to rigorously distinguish between mental and physical fatigue. Guo et al. [7] achieved 88.6% accuracy classifying flight fatigue but acknowledged that physical and mental fatigue were not separately investigated, with a general fatigue classification adopted instead. Shen et al. [27] and Pickering et al. [23] emphasised that mental fatigue, physical fatigue, and boredom have distinct mechanisms but are frequently conflated. The present study addresses this by excluding physical activity phases from the training data and validating through VAS scores that the labelled fatigued state corresponds specifically to mental fatigue.

3 Methodology

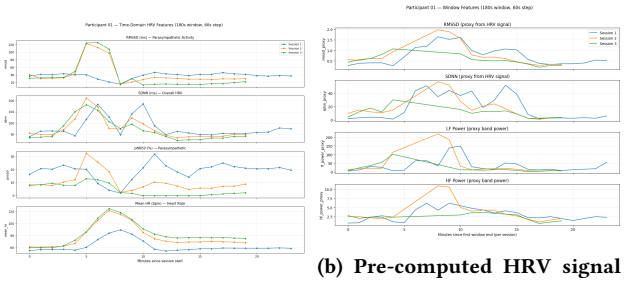
3.1 Dataset

The present study employs the publicly available FatigueSet dataset [13], comprising 12 participants (9 male, 3 female, ages 21–40, mean age 30.75 years), across 3 sessions each, totalling 36 sessions. The study followed a within-subjects design, with session order counterbalanced across participants using a balanced Latin Square design to control for order effects. Each session followed a structured protocol in which participants completed a 3-minute baseline resting period, a fatigue measurement point (M1: VAS survey, choice reaction time task, and n-back task), a physical activity phase on a treadmill at low (5 km/h, 3.2 METs), medium (7 km/h, 5.3 METs), or high intensity (9 km/h, 8.8 METs), a second measurement point (M2), an approximately 15-minute cognitive dual task involving letter and number identification designed to induce mental fatigue, and a final measurement point (M3). The CRT and n-back tasks were used as measures of fatigability, capturing processing speed and working memory performance respectively, with changes relative to baseline (M1) used as indicators of cognitive decline. Although the full dataset includes additional wearable devices such as a Muse S EEG headband and an earable prototype, this study focuses on data recorded from the Zephyr BioHarness 3.0 chest strap, which provides beat-to-beat RR intervals, and the Empatica E4 wristband, which records wrist skin temperature at 4 Hz.

3.2 Data Preprocessing

3.2.1 RR Interval Selection. The FatigueSet dataset provides both a pre-computed HRV summary signal and raw beat-to-beat RR intervals. Initial analysis compared features derived from both sources. Features computed from the pre-computed summary signal produced values with extremely limited dynamic range, for example, RMSSD values ranged from 0.3 to 2.0 (unitless), with differences between baseline and fatigued states as small as 0.1, making it difficult for classifiers to distinguish between physiological states. In contrast, features computed from raw RR intervals produced RMSSD values of 20-150 ms with clear separation of approximately 20 ms between baseline and fatigued states. The resting baseline values of approximately 20-75 ms are consistent with published short-term norms of 19-75 ms for healthy adults [26]. Figure 1 illustrates this comparison. Consequently, all HRV features in this study were computed directly from raw RR intervals, in accordance with the recommendations of the Task Force of the European Society of Cardiology [16].

3.2.2 RR Interval Cleaning. Raw RR intervals were preprocessed to remove physiologically implausible values and ectopic beats. Intervals outside the range of 300-2000 ms (corresponding to heart rates of approximately 30-200 bpm) were excluded. Successive RR intervals differing by more than 300 ms were also removed to eliminate ectopic beats and motion artifacts. Windows containing fewer than 100 cleaned RR intervals after filtering were discarded to ensure sufficient data for reliable feature computation.



(a) Raw RR interval features

(b) Pre-computed HRV signal (proxy)

Figure 1: Comparison of RMSSD computed from pre-computed HRV summary (right) versus raw RR intervals (left) for Participant 01. The pre-computed signal produces values with insufficient dynamic range (0.3-2.0, unitless) to distinguish between sessions, while raw RR intervals produce physiologically interpretable values (20-150 ms) with clear separation between baseline and fatigued states.

3.2.3 Skin Temperature Preprocessing. Wrist skin temperature data, recorded at 4 Hz by the Empatica E4, were resampled to 1 Hz using mean aggregation to reduce high-frequency noise while preserving the slow-changing thermal dynamics relevant to fatigue detection.

3.3 Feature Extraction

Features were extracted using a sliding window approach with a 180-second window and 60-second step size (66.7% overlap). The 180-second window was selected to ensure reliable computation of frequency-domain HRV features, which require a minimum observation period of 2-3 minutes for stable spectral estimates [16]. As shown in Figure 2, SVM classification performance increased monotonically with window size, with the 180-second window achieving the highest F1 score, suggesting that longer windows provide more stable spectral estimates that benefit classification. The 60-second step size was empirically validated by comparing six step sizes (15s, 30s, 60s, 90s, 120s, and 180s), as shown in Figure 3. The 60-second step yielded the highest SVM F1 score (0.784), outperforming both smaller step sizes which produced highly redundant windows (e.g., 15s step: 92% overlap, F1 = 0.750) and larger step sizes which provided insufficient training data (e.g., 180s step: 0% overlap, F1 = 0.727). These results informed the selection of the 180-second window and 60-second step size adopted throughout this study.

3.3.1 HRV Features. Twenty-two HRV features were computed from the cleaned RR intervals within each window, spanning two domains as summarised in Table 1. For frequency-domain analysis, RR intervals were resampled to a uniform 4 Hz signal using cubic spline interpolation. Welch's method was then applied using a 256-point window with 50% overlap to estimate the power spectral density. Power was integrated across the very low frequency (VLF: 0.003-0.04 Hz), low frequency (LF: 0.04-0.15 Hz), and high frequency (HF: 0.15-0.40 Hz) bands in accordance with established standards [16]. Natural logarithm transforms were additionally applied to all absolute power measures, following the convention of expressing HRV indices in log units to achieve a more normal distribution [26].

3.3.2 Temperature Features. Six skin temperature features were computed per window: mean temperature, linear slope (estimated via linear regression, in °C per second), standard deviation (variability), range, minimum value, and change from baseline. Prior to feature extraction, the raw temperature signal was resampled to 1 Hz using mean aggregation. Windows containing fewer than 10 valid temperature readings after resampling were discarded. The change from baseline was calculated by subtracting the first window's mean temperature within each participant-session from all subsequent windows, thereby removing inter-individual differences in absolute skin temperature. This assumes a stable resting baseline is available at session onset, a condition satisfied by the FatigueSet protocol but not guaranteed in unstructured real-world settings. Features were extracted using the same 180-second sliding window with a 60-second step as the HRV pipeline.

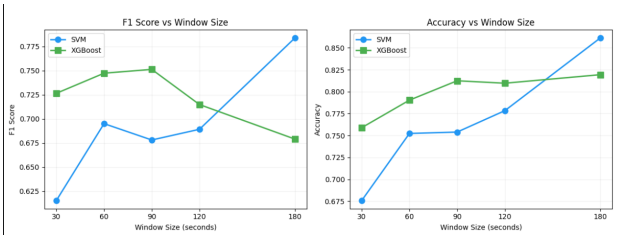


Figure 2: Effect of window size on SVM and XGBoost classification performance with a fixed 60-second step size. SVM F1 score increases monotonically with window size, with the 180-second window achieving the highest performance (F1=0.778). Data availability decreases as window size increases due to fewer complete windows fitting within each session.

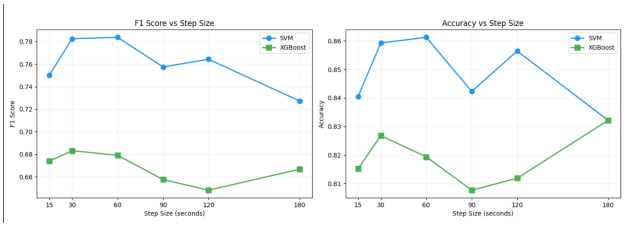


Figure 3: Effect of step size on SVM and XGBoost classification performance with a fixed 180-second window. A 60-second step (66.7% overlap) achieved the highest SVM F1 score, balancing training data volume against inter-window redundancy.

Table 1: Summary of extracted features (28 total).

Domain	Features	Count
Time-domain	mean RR, SDNN, RMSSD, pNN50, mean/min/max HR, SDHR, RR range, CV_RR, triangular index	11
Frequency-domain	VLF, LF, HF, total power, LF/HF ratio, normalised LF, normalised HF, ln(VLF), ln(LF), ln(HF), ln(total power)	11
Temperature	mean, slope, std, range, min, change from baseline	6

3.4 Labelling Strategy

Windows were labelled using experimental phase timestamps from the event markers file. Windows occurring between `start_baseline`

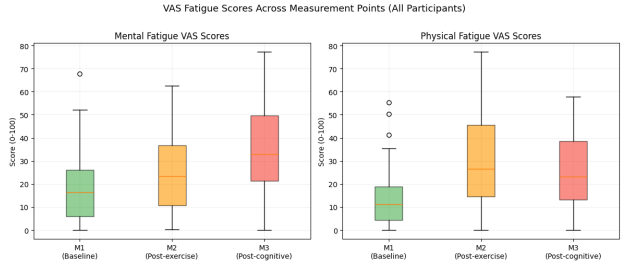


Figure 4: VAS fatigue scores across measurement points. Mental fatigue (left) increased significantly from M1 to M3 ($p < 0.001$), while physical fatigue (right) peaked at M2 post-exercise and decreased by M3, suggesting that the labelled fatigued state reflects mental rather than physical fatigue.

and `start_activity` were labelled as *baseline* (class 0, not fatigued). Windows occurring from `start_fatigue` (the beginning of the cognitive dual task) until `end_session` were labelled as *fatigued* (class 1). Windows during the physical exercise and recovery phases were excluded from the dataset, as these periods contain physiological responses to physical exertion that could confound mental fatigue detection. This exclusion was validated empirically; including exercise and recovery windows as "not fatigued" reduced SVM F1 from 0.778 to 0.511, suggesting that physical activity introduces confounding physiological signals.

Labels were further validated using Visual Analogue Scale (VAS) mental fatigue scores collected at three measurement points during each session, as shown in Figure 4. Mental fatigue VAS increased significantly from baseline (M1: 18.4 ± 15.5) to post-cognitive task (M3: 36.4 ± 21.9, $p < 0.001$, Wilcoxon signed-rank test), suggesting that the cognitive dual task successfully induced mental fatigue. Importantly, physical fatigue VAS peaked at M2 (post-exercise: 24.5) and decreased by M3 (20.1), suggesting that the labelled fatigued state at M3 reflects mental rather than physical fatigue. The final dataset comprised 402 labelled windows (275 baseline, 127 fatigued) from 12 participants across 36 sessions.

3.5 Statistical Analysis

To determine which features differ significantly between baseline and fatigued states, Mann-Whitney U tests were applied independently to each of the 22 HRV features and 6 skin temperature features. The Mann-Whitney U test was selected as a non-parametric alternative to the independent samples t-test, as it makes no assumptions about the normality of the underlying distributions, which is appropriate given the small sample size of 12 participants and the windowed nature of the data. Effect sizes were computed using rank-biserial correlation. A significance threshold of $p < 0.05$ was applied. It should be noted that the 402 windows were drawn from the same participants across sessions. The reported p-values should therefore be interpreted as indicative of group-level trends rather than strictly independent statistical tests. Features were additionally ranked by effect size to identify those with the strongest discriminative signal beyond statistical significance alone.

To further characterise the progression of fatigue, a three-class analysis was also conducted distinguishing baseline, mild fatigue (post-exercise recovery phase), and fatigued (post-cognitive task) states. A Kruskal-Wallis omnibus test was first applied to each feature to determine whether any significant differences existed across the three groups. For features showing a significant omnibus result, pairwise Mann-Whitney U tests were conducted between all three class pairs: baseline versus mild fatigue, mild fatigue versus fatigued, and baseline versus fatigued, with effect sizes computed for each comparison and presented in Figure 7.

Table 2: Final hyperparameter configurations.

Model	Configuration
SVM	$C = 5.0$, RBF kernel, `gamma=scale`
Random Forest	100 trees, max depth 10, min leaf 2
XGBoost	50 trees, max depth 3, lr=0.1, `scale_pos_weight=2.0`
Logistic Reg.	L1 penalty, $C = 0.1$

3.6 Classification Models

3.6.1 Traditional Machine Learning. Four classifiers were evaluated: Support Vector Machine (SVM) with radial basis function (RBF) kernel, Random Forest (RF), XGBoost, and Logistic Regression. All models used class weighting (`class_weight="balanced"` or `scale_pos_weight`) to account for the 2.2:1 class imbalance between baseline and fatigued windows. SVM and Logistic Regression features were standardised using z-score normalisation fitted on training data only to prevent data leakage. Hyperparameters were selected through systematic evaluation within the LOSO framework; final configurations are listed in Table 2.

Four feature subsets were compared: all 28 features, HRV-only (22 features), temperature-only (6 features), and a reduced set of 13 features identified through feature ablation. SMOTE oversampling [4] was additionally evaluated to address class imbalance.

3.6.2 Deep Learning. A Multi-Layer Perceptron (MLP) with three hidden layers (64, 32, and 16 neurons), batch normalisation [12], and dropout rates of 0.3, 0.3, and 0.2 respectively [28] was applied to the same extracted features as the traditional classifiers. Additionally, three deep learning architectures were applied directly to raw RR intervals to evaluate whether end-to-end learning could bypass manual feature extraction, namely a 1D Convolutional Neural Network (CNN), a Long Short-Term Memory network (LSTM), and a hybrid CNN-LSTM architecture. Raw RR sequences were padded or truncated to a fixed length of 250 beats per window, with shorter sequences padded using the window mean RR value as a neutral fill. Windows containing fewer than 50 clean beats were excluded. All deep learning models were trained using the Adam optimiser [14] with binary cross-entropy loss, with class weights set proportional to the inverse class frequency to address the 2.2:1 class imbalance. Early stopping with a patience of 5 epochs was applied to prevent overfitting. Due to the stochastic nature of neural network training, all deep learning results were averaged over five runs with fixed random seeds (42-46) and reported as mean ± standard deviation.

3.6.3 Autoencoder-Based Approaches. Two autoencoder-based approaches were additionally evaluated. The first used an autoencoder (encoder: input → 16 → bottleneck; decoder: bottleneck → 16 → input) with bottleneck dimensions of 5 and 7 to compress the 13 extracted features into a lower-dimensional representation, which was subsequently passed to an SVM or MLP classifier. The autoencoder was trained on the training fold only within each LOSO iteration to prevent data leakage. The second approach trained an autoencoder exclusively on baseline windows, treating the mean squared reconstruction error (MSE) as an anomaly

score whereby windows exceeding an optimised threshold were classified as fatigued. The detection threshold was selected by maximising F1 score over percentiles of the error distribution.

3.7 Feature Ablation

To identify a reduced feature subset, a systematic ablation study was conducted by removing one feature at a time from the full 28-feature set and measuring the resulting change in F1 score under LOSO cross-validation. Features whose removal consistently degraded performance across multiple models were retained, while features whose removal had negligible or positive impact were discarded. Additionally, modality-specific subsets were tested: HRV features only (22 features), temperature features only (6 features), and the combined set (28 features). This analysis identified a reduced subset of 13 features that outperformed the full set, comprising 5 temperature features (*temp_change_from_baseline*, *temp_range*, *temp_std*, *temp_slope*, *temp_min*) and 8 HRV features (*sdhr*, *cv_rr*, *triangular_index*, *max_hr*, *vlf_power*, *mean_rr*, *min_hr*, *pnn50*).

3.8 Evaluation

All models were evaluated using Leave-One-Subject-Out (LOSO) cross-validation, in which the model was trained on data from 11 participants and tested on the held-out participant, rotating through all 12. This ensured that classification performance reflected genuine cross-subject generalisation, as the model never encountered any data from the test participant during training. To prevent data leakage, all feature scaling was fitted exclusively on the training fold and applied to the test fold. Performance was assessed using accuracy, F1 score, precision, recall, and confusion matrix, with F1 score as the primary metric due to the 2.2:1 class imbalance between baseline and fatigued windows. Per-participant F1 scores were additionally recorded to assess inter-individual variability in model performance across subjects.

4 Results

4.1 Data Validation

Prior to classification, the chronological ordering of labels was verified across all 36 participant-sessions to ensure that no session exhibited label flip-flopping, defined as fatigued windows being followed by baseline windows within the same session. This verification confirmed that the labelling strategy produces temporally coherent annotations, in which baseline precedes fatigue rather than fluctuating randomly. As shown in Appendix A, all 36 sessions follow the expected temporal progression: baseline → exercise (excluded) → recovery (excluded) → fatigued. Zero sessions out of 36 exhibited flip-flopping, validating that the phase timestamps derived from the event markers file produce labels consistent with the experimental protocol.

4.2 Statistical Feature Analysis

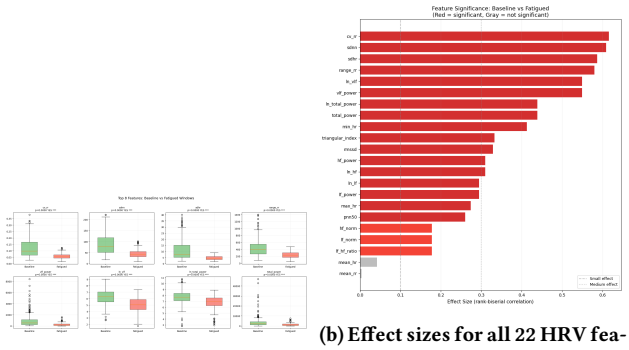
4.2.1 Label Validation. Prior to feature analysis, the labelling strategy was validated using VAS mental fatigue scores collected at three measurement points across all 36 sessions. As shown in Figure 4, mental fatigue increased significantly from baseline (M1: 18.4 ± 15.5) to post-cognitive task (M3: 36.4 ± 21.9, $p < 0.001$, Wilcoxon signed-rank test), suggesting that the 15-minute dual cognitive task successfully induced mental fatigue. A smaller but also significant increase was observed from M1 to post-exercise (M2: 24.5 ± 17.6, $p = 0.002$), reflecting the moderate cognitive demand of the measurement tasks themselves. Crucially, physical fatigue peaked at M2 and declined by M3, suggesting that windows labelled as fatigued at M3 reflect mental rather than physical fatigue. The final labelled dataset comprised 275 baseline windows and 127 fatigued windows across 12 participants and 36 sessions.

4.2.2 Binary Analysis: Baseline vs Fatigued. Mann-Whitney U tests revealed that 26 of 28 features differed significantly ($p < 0.05$) between baseline and fatigued states, though these p-values should be interpreted with caution given the correlated nature of overlapping windows from the same participants. All 6 temperature features and 20 of 22 HRV features showed significant differences. The two non-significant HRV features were mean heart rate ($p = 0.498$) and mean RR interval ($p = 0.955$). This is likely because baseline windows include the M1 measurement period, during which participants performed the choice reaction time and n-back tasks, elevating heart rate to levels comparable to those observed during the fatigued phase. As a result, the mean level of heart rate does not reliably distinguish the two states, whereas variability-based features remain discriminative because cognitive task performance suppresses autonomic flexibility even when mean heart rate is similar.

The complete effect size ranking is shown in Figure 5. The top-ranked features were all measures of variability rather than central tendency. *cv_rr* achieved the highest effect size ($r = 0.616$, $p < 0.001$), followed by SDNN ($r = 0.609$) and SDHR ($r = 0.587$). *cv_rr* normalises beat-to-beat variability relative to mean heart rate, making it robust to inter-individual differences in absolute RR values and more sensitive to fatigue-induced autonomic suppression than unnormalised measures such as SDNN or RMSSD. SDNN and SDHR capture overall variability across all time scales, and both collapsed substantially during fatigue (SDNN: 87.2 → 45.0 ms; SDHR: 11.1 → 4.6 bpm), reflecting a broad reduction in autonomic flexibility under sustained cognitive load.

Frequency-domain features showed consistently strong effects. VLF power ($r = 0.550$) decreased from 952.1 to 261.3 ms^2, and total spectral power ($r = 0.439$) fell from 3706.5 to 1450.9 ms^2, indicating a broad suppression of autonomic activity across all frequency bands. HF power ($r = 0.310$), reflecting parasympathetic vagal activity, decreased from 1229.3 to 327.5 ms^2, consistent with vagal withdrawal during cognitive stress. The LF/HF ratio increased from 3.83 to 4.58 ($r = 0.178$, $p = 0.004$), indicating a relative shift towards sympathetic dominance, though with a smaller effect than the absolute power measures, as normalised ratios are less sensitive to the overall reduction in autonomic activity. RMSSD ($r = 0.329$) and pNN50 ($r = 0.260$) both decreased significantly, reflecting reduced parasympathetic modulation at the beat-to-beat level. An exception was *min_hr* ($r = 0.413$), which increased from 61.2 to 71.2 bpm during fatigue, suggesting that the lowest instantaneous heart rates were elevated, consistent with reduced vagal slowing of the heart.

All six skin temperature features reached statistical significance, as shown in Figure 6. The strongest discriminators were *temp_range* ($r = 0.477$, $p < 0.001$), which decreased from 0.44°C to 0.21°C, and *temp_std* ($r = 0.468$, $p < 0.001$), which decreased from 0.125 to 0.056°C, indicating that temperature became notably more stable during the fatigued state. The temperature slope (*temp_slope*) increased slightly from −0.0004 to +0.0001°C/s ($r = 0.262$, $p < 0.001$), suggesting a shift from cooling to warming during the cognitive task. The temperature change from baseline



(a) Box plots for the top 8 HRV features by effect size.

(b) Effect sizes for all 22 HRV features ranked by discriminative strength.

Figure 5: Statistical analysis of HRV features comparing baseline and fatigued windows. Red bars indicate statistically significant features ($p < 0.05$), grey bars indicate non-significant features.

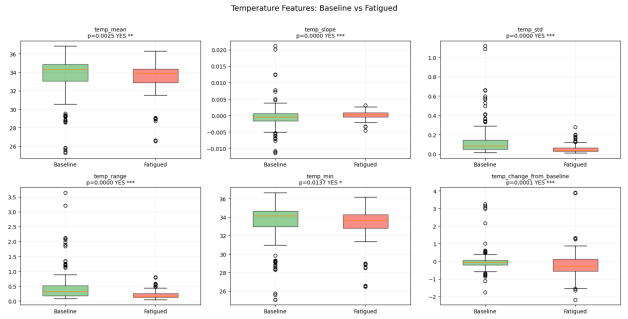


Figure 6: Box plots for the top 8 temperature features by effect size.

decreased further during fatigue ($r = 0.258$, $p < 0.001$), indicating a gradual skin temperature decline relative to session start. Mean temperature showed a smaller but significant decrease from 33.74°C to 33.37°C ($r = 0.193$, $p = 0.003$), consistent with sympathetically mediated peripheral vasoconstriction during sustained cognitive effort.

4.2.3 Three-Class Analysis: Baseline vs Mild Fatigue vs Fatigued. To characterise the progression of fatigue across the session, a three-class analysis was conducted distinguishing baseline (M1, VAS ≈ 18), mild fatigue (post-exercise recovery, VAS ≈ 24), and fatigued (post-cognitive task, M3, VAS ≈ 36) states. It should be noted that the mild fatigue class corresponds to the post-exercise recovery period, meaning its physiological signal reflects a mixture of residual exercise effects and early cognitive fatigue rather than a purely incremental mental fatigue state.

Kruskal-Wallis omnibus tests revealed that 20 out of 22 HRV features showed significant differences across the three groups, with only *mean_rr* ($p = 0.114$) and *mean_hr* ($p = 0.180$) remaining non-significant, consistent with the binary analysis. Pairwise Mann-Whitney U tests, visualised in Figure 7, revealed an important asymmetry in how fatigue progresses across the session. Seventeen out of 22 features significantly distinguished baseline from mild fatigue, indicating that the physiological changes induced by physical exercise and early recovery were already sufficient to substantially alter the autonomic state. In contrast, only 3 features (*sdhr*, *cv_rr*, and *max_hr*) significantly distinguished

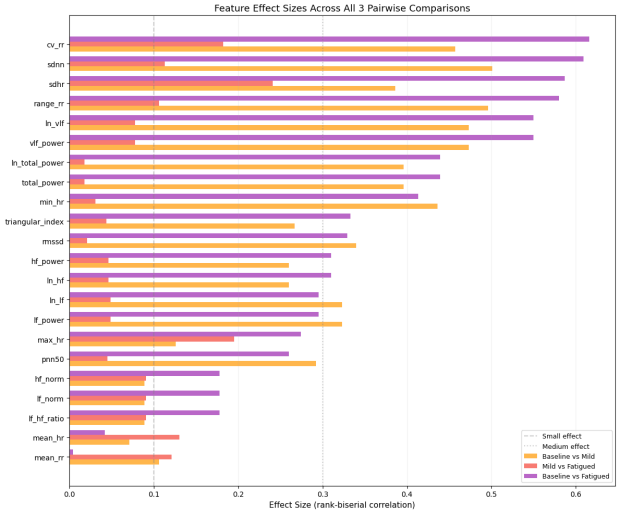


Figure 7: Effect sizes (rank-biserial correlation) for all 22 HRV features across three pairwise comparisons: baseline vs mild fatigue (orange), mild fatigue vs fatigued (red), and baseline vs fatigued (purple). Dashed vertical lines indicate small (0.1) and medium (0.3) effect size thresholds. Features are ranked by baseline vs fatigued effect size.

mild fatigue from the fully fatigued state. This suggests that the largest physiological transition occurs at the baseline-to-mild-fatigue boundary, driven primarily by the acute cardiovascular response to exercise, while the subsequent transition from mild to full mental fatigue produces comparatively subtler autonomic changes that most features cannot detect.

These findings have practical implications for the classification task. The overlap between physiological signatures of post-exercise recovery and sustained mental fatigue, evidenced by only 3 features distinguishing these two states, justifies the binary labelling strategy adopted in this study, in which exercise and recovery windows are excluded entirely. Only *sdhr*, *cv_rr*, and *max_hr* distinguished all three pairwise comparisons, reinforcing the suggestion that heart rate variability measures are the most informative physiological dimension for tracking fatigue as a continuous progression.

4.3 Baseline Model Performance

Four classifiers were evaluated on the full 28-feature set under LOSO cross-validation to establish baseline performance prior to feature optimisation. Results are summarised in Table 3 and Figure 8.

XGBoost achieved the highest accuracy (0.843) while SVM and XGBoost achieved comparable F1 scores (0.699 and 0.700 respectively), with Random Forest (F1=0.610) and Logistic Regression (F1=0.649) performing comparatively lower. The consistent gap between accuracy and F1 across all models reflects the 2.2:1 class imbalance, whereby a classifier that simply predicts the majority baseline class more frequently will achieve high accuracy without necessarily detecting fatigued windows well. F1 score is therefore the more informative metric and is used as the primary comparison criterion throughout.

The confusion matrix for XGBoost, provided in Appendix D, reveals that 256 of 275 baseline windows were correctly classified (93.1%), while only 71 of 127 fatigued windows were correctly

Table 3: Baseline model performance on full 28-feature set (LOSO CV).

Model	Accuracy	F1	Precision	Recall
Random Forest	0.799	0.610	0.709	0.535
SVM	0.807	0.699	0.644	0.763
Logistic Regression	0.766	0.649	0.579	0.737
XGBoost	0.843	0.700	0.798	0.623



Figure 8: Baseline model comparison across four classifiers trained on all 28 features under LOSO cross-validation. XGBoost achieved the highest accuracy while SVM and XGBoost achieved comparable F1 scores.

detected (55.9%). This pattern of high baseline precision but moderate fatigued recall is consistent across all models and reflects the difficulty of generalising fatigue detection to unseen participants under LOSO cross-validation.

Notable differences in the precision-recall trade-off were observed across models. Random Forest exhibited the highest precision (0.709) but the lowest recall (0.535), indicating a conservative classifier that predicts fatigue only when highly confident but misses a large proportion of fatigued windows. Logistic Regression showed the inverse, with lower precision (0.579) but higher recall (0.737), suggesting a tendency to over-predict the fatigued class. SVM balanced these most effectively, with precision 0.644 and recall 0.763, while XGBoost achieved the best accuracy-F1 combination with precision 0.798 and recall 0.623.

Per-participant results for XGBoost revealed substantial inter-individual variability. Participants P02, P03, and P05 achieved near-perfect F1 scores (1.000, 0.947, and 0.941 respectively), while P01, P07, P09, and P12 showed considerably lower F1 scores (0.480, 0.462, 0.500, and 0.444). This variability suggests that physiological fatigue signatures are more consistent and generalisable for some participants than others, and motivates the need for further feature and model optimisation to improve cross-subject robustness.

4.4 Unsupervised Feature Space Analysis

Prior to feature optimisation, the viability of unsupervised classification was investigated alongside an analysis of the feature space structure, to determine whether fatigue states could be detected without labelled training data.

Principal Component Analysis (PCA) reduced the 28 features to 2 components explaining 38.2% and 15.5% of variance respectively (53.7% combined), with 11 components required to capture 95% of total variance (Appendix B). This indicates that the feature space contains significant redundancy, while 28 features were extracted, only 11 transformed dimensions are needed to represent nearly all meaningful variation in the data, with the

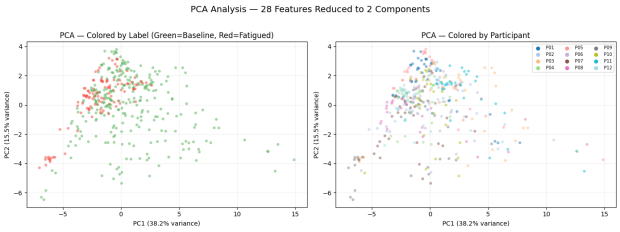


Figure 9: PCA projection coloured by class label (left) and participant identity (right). PC1 explains 38.2% and PC2 explains 15.5% of total variance. Substantial class overlap and participant-specific clustering are both visible.

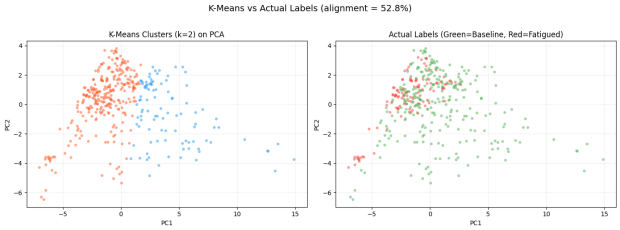


Figure 10: K-Means ($k = 2$) cluster assignment versus actual labels on the PCA projection. Alignment of 52.8% with true labels suggests that unsupervised clustering cannot reliably separate baseline from fatigued states in this feature space.

remaining dimensions contributing primarily noise. Training models on all 28 features therefore risks fitting to these noisy dimensions, which may lead to overfitting whereby the model learns patterns specific to training participants that do not generalise to unseen individuals. This provides additional motivation for feature reduction through ablation, discussed in Section 4.5.

The 2D projection coloured by class label (Figure 9) reveals substantial overlap between baseline and fatigued windows, indicating that the fatigue signal is distributed across multiple dimensions and cannot be captured by a simple low-dimensional projection. The projection coloured by participant identity reveals partially distinct participant clusters, suggesting the presence of strong inter-individual physiological differences that motivated the LOSO evaluation strategy.

K-Means clustering with $k = 2$ was evaluated as an unsupervised classifier by measuring alignment between discovered clusters and true labels. The best cluster assignment achieved only 52.8% accuracy, marginally above random chance (50%), as shown in Figure 10. The confusion matrix revealed that 179 of 275 baseline windows were misassigned, while 110 of 127 fatigued windows were correctly grouped, indicating that the clusters discovered by K-Means do not correspond to the fatigue-relevant structure in the data. This suggests that unsupervised clustering alone is insufficient for reliable mental fatigue detection from HRV and temperature features, and that supervised learning with ground-truth labels may be necessary, at least for the methods evaluated here. Notably, this near-chance alignment is not merely a negative result; it reveals that mental fatigue produces a physiologically *more uniform* signature than the alert baseline state, which is precisely why reconstruction-based anomaly detection also failed to detect it reliably (Section 4.9): a signal characterised by reduced variance is inherently difficult to flag as anomalous.

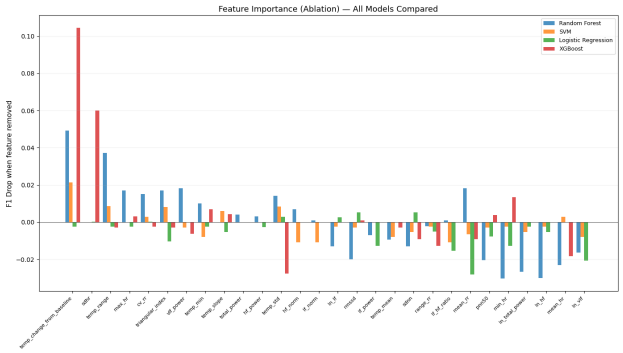


Figure 11: Cross-model feature ablation comparison. Features are ranked by average F1 drop across all four models. *temp_change_from_baseline* and *sdhr* are the most consistently important features across models.

4.5 Feature ablation

Feature ablation was conducted by removing one feature at a time from the full 28-feature set and measuring the resulting change in F1 score across all four baseline models under LOSO cross-validation. Individual per-model ablation plots are provided in Appendix C, and the combined cross-model comparison is shown in Figure 11.

The most consistently important feature across all models was *temp_change_from_baseline*, whose removal caused the largest F1 drop in Random Forest (ΔF1= +0.049), SVM (ΔF1= +0.021), and XGBoost (ΔF1= +0.104). This suggests that cumulative temperature drift relative to a participant's own baseline is the single most discriminative feature in the dataset, likely because it captures the progressive thermoregulatory response to sustained cognitive load while normalising inter-individual differences in absolute skin temperature. *sdhr* was the second most important feature for XGBoost (ΔF1= +0.060) and also ranked highly for Random Forest, reinforcing the finding from the statistical analysis that heart rate variability measures are the strongest HRV discriminators. *temp_range* and *temp_std* were consistently important across RF and SVM, while Logistic Regression relied more heavily on time-domain HRV features such as *sdnn* and *rmssd*.

A notable finding is the disagreement between models on which features are harmful versus helpful. Many features that are statistically significant (Section 4.2) showed negative F1 drops when removed from SVM and Logistic Regression, indicating that their inclusion actively hurt performance despite being individually discriminative. This is consistent with multicollinearity among HRV features, for example, *ln_vlf*, *ln_lf*, *ln_hf*, and *ln_total_power* are all correlated transformations of the same spectral power measures, and including all of them adds noise rather than signal for linear classifiers.

Based on the ablation results, features were retained if their removal caused a positive F1 drop in at least two out of four models. This produced a reduced subset of 13 features: 5 temperature features (*temp_change_from_baseline*, *temp_range*, *temp_std*, *temp_slope*, *temp_min*) and 8 HRV features (*sdhr*, *cv_rr*, *triangular_index*, *max_hr*, *vlf_power*, *mean_rr*, *min_hr*, *pnn50*). The Random Forest Gini feature importance, provided in Appendix J, corroborates this selection, with the features mentioned above occupying the top six positions.

4.6 Model Improvements

Following feature ablation, a series of optimisation experiments were conducted to improve upon the baseline performance. Four improvements were investigated: feature subset selection, hyperparameter tuning, SMOTE oversampling, and per-participant normalisation.

4.6.1 Feature Subset Comparison. The impact of feature selection was evaluated by comparing four subsets under LOSO cross-validation using XGBoost, with full results provided in Appendix L. The reduced 13-feature set outperformed the full 28-feature set (F1=0.738 vs F1=0.700), suggesting that removing redundant and correlated features improves generalisation. Critically, HRV-only features (22 features) performed substantially worse (F1=0.376), while temperature-only features (6 features) achieved a moderate F1 of 0.570. This suggests that neither modality alone may be sufficient for reliable fatigue detection in this dataset, and that combining HRV and temperature features is beneficial. The poor performance of HRV-only under LOSO suggests that HRV features are more susceptible to inter-individual variability than temperature features, which benefit from the participant-relative normalisation provided by *temp_change_from_baseline*.

4.6.2 Hyperparameter Tuning. Systematic hyperparameter search was conducted for all four models using the reduced 13-feature set. The best configurations, as reported in Table 4, yielded improvements across all models compared to their default baselines. SVM improved most substantially, from F1=0.699 (baseline) to F1=0.778 (tuned, $C = 5.0$, RBF kernel), representing a ΔF1= +0.079 improvement. Random Forest improved from F1=0.610 to F1=0.700, while XGBoost improved from F1=0.700 to F1=0.740. Logistic Regression showed the least improvement (F1=0.644), consistent with its inability to model the non-linear decision boundary in the feature space. The overall comparison between baseline and improved models is provided in Appendix K.

4.6.3 SMOTE Oversampling. SMOTE oversampling was applied within the training fold of each LOSO iteration to address the 2.2:1 class imbalance. Results were mixed across models. RF improved by ΔF1= +0.031 (0.663 → 0.694) and SVM improved by ΔF1= +0.013 (0.741 → 0.754), suggesting that synthetic oversampling helped these models better detect the minority fatigued class. XGBoost, however, degraded by ΔF1= −0.026 (0.738 → 0.712), likely because XGBoost already handles class imbalance internally via `scale_pos_weight` and the additional synthetic samples introduced noise rather than useful signal.

4.6.4 Final Best Configuration. The complete comparison across all configurations is shown in Figure 12. The best overall configuration was SVM with the reduced 13-feature set and tuned hyperparameters ($C = 5.0$, RBF kernel, `gamma=scale`), achieving F1=0.778 and accuracy=0.863 under LOSO cross-validation. This represents a substantial improvement over the baseline SVM (F1=0.699, ΔF1= +0.079), baseline XGBoost (F1=0.700, ΔF1= +0.078), and baseline Random Forest (F1=0.610, ΔF1= +0.168). Logistic Regression remained the weakest classifier throughout all experiments (best F1=0.644), suggesting that the fatigue classification boundary in this feature space is non-linear and cannot be adequately modelled by a linear classifier.

The confusion matrix for the best configuration is provided in Appendix E. Of 275 baseline windows, 257 were correctly classified as baseline and 18 were incorrectly predicted as fatigued

Table 4: Final hyperparameter configurations after tuning on the reduced 13-feature set (LOSO CV).

Model	Configuration	Base F1	Tuned F1
SVM	C=5.0, RBF, `gamma=scale`	0.699	0.778
Random Forest	100 trees, depth 10, leaf 2	0.610	0.700
XGBoost	50 trees, depth 3, lr=0.1	0.700	0.740
Logistic Reg.	L1, C=0.1	0.649	0.644

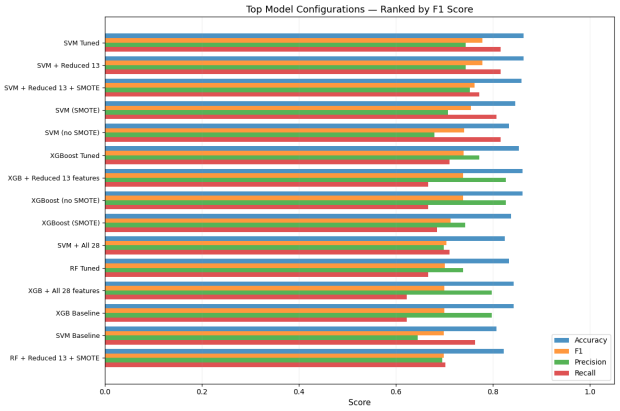


Figure 12: All model configurations ranked by F1 score. SVM with reduced 13 features and tuned hyperparameters achieves the best F1 of 0.778.

(false positive rate=6.5%). Of 127 fatigued windows, 103 were correctly identified as fatigued and 24 were missed as baseline (false negative rate=18.9%). The asymmetry between false positive and false negative rates reflects the precision-recall trade-off inherent in the 2.2:1 class imbalance. The model correctly identifies the majority of fatigued windows (recall=0.816) while maintaining reasonable precision (0.744), making it more suitable for fatigue detection applications in which missing a fatigued state is more costly than a false alarm.

Per-participant results revealed substantial inter-individual variability. Eight of 12 participants achieved F1 $\geq$ 0.733, with P03 achieving perfect classification (F1=1.000), and P04, P05, P06, P10, and P11 all achieving F1 $\geq$ 0.833. These participants likely exhibit more consistent and generalisable physiological fatigue signatures that the model can reliably learn from other participants under LOSO. The four lower-performing participants (P07: F1=0.462, P08: F1=0.500, P09: F1=0.688, P12: F1=0.593) showed considerably weaker performance, suggesting that their fatigue-related physiological responses deviate from the population-level patterns captured in the training data. This variability is consistent with the findings of the statistical analysis, in which individual participants showed markedly different HRV baseline values and fatigue-induced changes, and highlights the fundamental challenge of cross-subject generalisation in physiological fatigue detection with a small participant pool of 12 individuals.

4.7 Labelling Strategy Validation

To validate the decision to exclude exercise and recovery windows from the training data, an expanded labelling strategy was evaluated by comparing two configurations:

- **Strategy A** (adopted): Baseline windows only as the negative class (274 windows, ratio 2.4:1)
- **Strategy B** (expanded): Baseline, exercise, and recovery windows combined as the negative class (531 windows, ratio 4.7:1)

Table 5: Strategy A (baseline only) vs Strategy B (expanded) classification results under LOSO cross-validation.

Strategy	Model	F1	Acc	Prec	Recall
A (baseline only)	SVM	0.778	0.863	0.744	0.816
A (baseline only)	RF	0.700	0.833	0.738	0.667
A (baseline only)	XGBoost	0.692	0.835	0.766	0.632
B (expanded)	SVM	0.511	0.754	0.393	0.728
B (expanded)	RF	0.349	0.780	0.365	0.333
B (expanded)	XGBoost	0.295	0.792	0.368	0.246

Results are summarised in Table 5, with the full comparison shown in Appendix I. Strategy B substantially degraded performance across all three models. SVM dropped from F1=0.778 to F1=0.511 (ΔF1= -0.267), RF from F1=0.700 to F1=0.349 (ΔF1= -0.352), and XGBoost from F1=0.692 to F1=0.295 (ΔF1= -0.398). Despite Strategy B having nearly twice as many training windows in the negative class, classification performance deteriorated severely, suggesting that data volume alone does not compensate for label noise.

This degradation occurs because exercise and recovery windows exhibit physiological states that overlap with mental fatigue in HRV feature space. Physical exercise elevates heart rate and suppresses HRV similarly to cognitive fatigue, while the recovery period involves a gradual return of autonomic function that spans a range of intermediate states. Including these windows in the negative class creates an ambiguous decision boundary whereby the model is simultaneously trained to treat exercise-like HRV patterns as both fatigued (when they occur during the fatigue phase) and not fatigued (when they occur during exercise), producing contradictory gradients that reduce classification confidence.

4.8 Deep Learning Results

4.8.1 Traditional Machine Learning vs Deep Learning. Six models were evaluated and compared against the best traditional ML configuration (SVM with reduced 13 features, F1=0.778) under LOSO cross-validation. Results are summarised in Table 6 and Figure 13.

SVM with hand-crafted features substantially outperformed all deep learning architectures. The MLP applied to all 28 features achieved F1=0.736, and the MLP on the reduced 13-feature set achieved F1=0.735, both competitive but consistently below SVM. The near identical performance between MLP on 28 and 13 features (ΔF1=0.001) suggests that the ablation-selected feature set retains essentially all information relevant to neural network classification, and that the additional 15 features contribute negligible discriminative signal.

The three raw RR interval architectures performed substantially worse. CNN-LSTM achieved the highest F1 among raw signal models (F1=0.502), followed by CNN (F1=0.432) and LSTM (F1=0.426). All three fell well below the feature-based models, with the gap between SVM and the best raw RR model exceeding ΔF1=0.276. The confusion matrices provided in Appendix G reveal that raw RR models produce substantially more false positives than the SVM. The CNN misclassified 127 of 274 baseline windows as fatigued, compared to only 32 for SVM, and LSTM misclassified 139. This suggests that without hand-crafted

Table 6: Deep learning model comparison under LOSO cross-validation (180-second window, 60-second step).

Model	Input	F1	Acc	Prec	Recall
SVM	13 features	0.778	0.863	0.744	0.816
MLP	28 features	0.736	0.830	0.677	0.807
MLP	13 features	0.735	0.833	0.687	0.789
CNN-LSTM	Raw RR (250)	0.502	0.619	0.428	0.606
CNN	Raw RR (250)	0.432	0.541	0.355	0.551
LSTM	Raw RR (250)	0.426	0.516	0.341	0.567

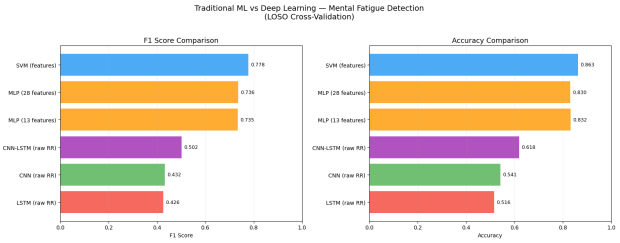


Figure 13: F1 score and accuracy comparison between traditional ML and deep learning models under LOSO cross-validation. SVM with hand-crafted features substantially outperforms all deep learning architectures. Feature-based MLP models are competitive but below SVM, while raw RR interval models perform poorly across both metrics.

features, these models cannot distinguish the two physiological states and instead develop a bias towards predicting fatigue, likely because the fatigued class has lower HRV variance, which presents as a more consistent pattern in the raw signal.

The superiority of traditional ML over deep learning in this context is consistent with the known limitations of data-hungry neural architectures on small physiological datasets. With only approximately 400 labelled windows and 12 participants under LOSO cross-validation, there is insufficient data for CNN and LSTM architectures to learn robust beat-to-beat temporal patterns that generalise to unseen individuals. The 22 hand-crafted HRV features used in this study encode decades of physiological knowledge about autonomic nervous system dynamics, effectively distilling the relevant information from 250 raw beats into compact discriminative features. This domain-specific compression gives the SVM a meaningful advantage over end-to-end models that must learn equivalent representations from scratch.

4.9 Autoencoder Results

4.9.1 Autoencoder Feature Reduction. An autoencoder with bottleneck dimensions of 3, 5, 7, and 10 was used to compress the 28-feature set into a lower-dimensional representation, subsequently passed to an SVM classifier. Results were compared against PCA baselines and the original feature set in Figure 14.

All autoencoder configurations underperformed the original 13-feature SVM (F1=0.778). The best autoencoder configuration (5-dimensional bottleneck) achieved F1=0.609, substantially below the hand-crafted feature baseline and even PCA with 10 components (F1=0.719). PCA consistently outperformed the autoencoder at equivalent dimensionality, suggesting that the linear variance structure captured by PCA is more informative for this classification task than the non-linear compressed representation learned by the autoencoder. With only 388 windows, the autoencoder likely learns a noisier and less generalisable compression than PCA, which has no parameters to overfit.

4.9.2 Autoencoder Combined with Deep Learning Across Window Sizes. To investigate whether combining autoencoders with neural networks or using smaller windows with more training data could improve performance, a multi-seed experiment was conducted across three window sizes (30s, 60s, 180s) comparing SVM, MLP, AE+SVM, AE+MLP, and AE anomaly detection. All deep learning results were averaged over five random seeds to account for stochastic training variability, and are reported as mean ± standard deviation. Results are shown in Figure 15.

SVM with hand-crafted features remained the best performing configuration across all window sizes (F1=0.784 at 180s). MLP at 180s achieved the closest result (F1=0.754±0.016), though still below SVM. AE+SVM and AE+MLP configurations ranged from F1=0.626 to F1=0.673 across window and bottleneck combinations, consistently below both SVM and plain MLP baselines. This suggests that autoencoder-based compression provides no benefit over direct feature use at this dataset scale, across the window sizes and classifiers evaluated.

Smaller windows did not rescue deep learning performance. At 30s and 60s, AE+MLP and AE+SVM achieved slightly higher F1 scores than at 180s for some configurations, but this improvement reflects the larger number of training windows available at shorter window sizes rather than a genuine improvement in feature quality. The standard deviations across seeds were also notably larger for AE+MLP configurations (e.g., AE(7)+MLP at 60s: F1=0.672±0.032), indicating instability in the learned representations that is absent for deterministic SVM.

AE anomaly detection showed a modest improvement at smaller windows (F1=0.613±0.007 at 30s vs F1=0.459±0.014 at 180s), which is the only case in which smaller windows helped. However, even the best anomaly detection result at 30s remained well below all supervised approaches, and the 30s window AE anomaly result (F1=0.613) coincidentally matches the SVM at 30s (F1=0.615), suggesting that the anomaly detector at shorter windows is simply benefiting from more training data rather than capturing meaningful fatigue structure.

Overall, these results suggest that for mental fatigue detection from HRV and skin temperature, domain-specific hand-crafted features combined with a kernel SVM may represent an effective approach at this dataset scale. Autoencoder-based compression and anomaly detection both failed to match supervised feature-based classification, and increasing data volume through shorter windows did not change this conclusion.

4.10 Summary of Results

Table 7 summarises the key configurations evaluated throughout this study, tracing the progression from baseline models to the final optimised configuration.

The central finding of this evaluation is that domain-specific feature engineering, rather than model architecture or hyperparameter tuning, is the primary performance driver in this dataset. Feature reduction from 28 to 13 features contributed as much to SVM's improvement as all subsequent hyperparameter tuning combined (ΔF1= +0.079 in both cases). Furthermore, raw RR interval models failed to close the gap with feature-based approaches even when training data was tripled via shorter windows, suggesting the bottleneck is the absence of physiological inductive bias rather than data volume. The best neural network

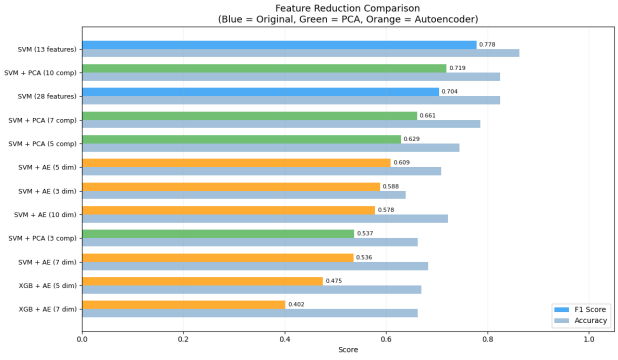


Figure 14: Feature reduction comparison: original features (blue), PCA (green), and autoencoder (orange). The 13-feature hand-crafted set outperforms all dimensionality reduction approaches. PCA consistently outperforms the autoencoder at equivalent dimensionality, suggesting insufficient data for non-linear compression to be beneficial.

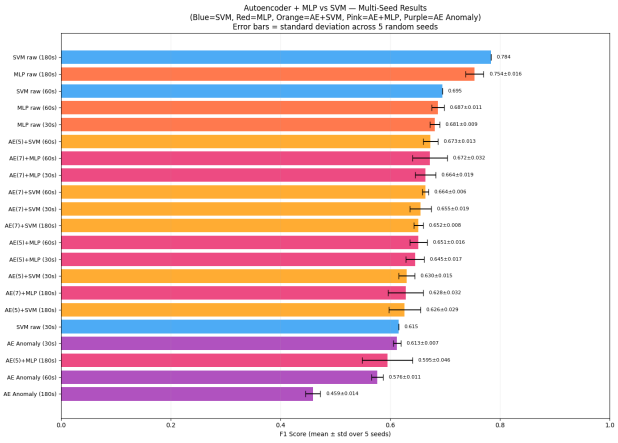


Figure 15: Multi-seed comparison (mean ± std over 5 seeds) of SVM, MLP, AE+SVM, AE+MLP, and AE anomaly detection across 30s, 60s, and 180s window sizes. SVM with hand-crafted features remains the best configuration. Error bars indicate standard deviation across 5 random seeds.

(MLP, F1=0.736) also remained below the tuned SVM despite operating on the same extracted features, consistent with the SVM's known sample efficiency advantage at this dataset scale. The final best configuration, SVM with the reduced 13-feature set ($C = 5.0$, RBF kernel, `gamma=scale`), achieved F1=0.778 and accuracy=0.863, representing a ΔF1= +0.168 improvement over the weakest baseline (RF, F1=0.610). Autoencoder-based compression and anomaly detection both failed to match hand-crafted feature performance, with the latter revealing a potential incompatibility between reconstruction-based outlier detection and the reduced-variability nature of mental fatigue. Prediction sequences across all 36 participant-sessions are provided in Appendix F, indicating that misclassifications are concentrated at the transition boundary between baseline and fatigued phases rather than distributed randomly throughout sessions.

Table 7: Progressive improvement of classification performance across all experiments (LOSO CV, 180s window, 60s step).

Stage	Configuration	F1	Acc	Prec	Recall
Baseline models (28 features, default hyperparameters)					
	Random Forest	0.610	0.799	0.709	0.535
	Logistic Reg.	0.649	0.766	0.579	0.737
	XGBoost	0.700	0.843	0.798	0.623
	SVM	0.699	0.807	0.644	0.763
Feature optimisation (13 features)					
	XGBoost	0.738	0.861	0.826	0.667
	SVM	0.778	0.863	0.744	0.816
Hyperparameter tuning (13 features)					
	RF (tuned)	0.700	0.833	0.738	0.667
	XGBoost (tuned)	0.740	0.853	0.771	0.711
	SVM (tuned)	**0.778**	**0.863**	0.744	0.816
Deep learning (180s window)					
	MLP (28 features)	0.736	0.830	0.677	0.807
	MLP (13 features)	0.735	0.833	0.687	0.789
	CNN-LSTM (raw RR)	0.502	0.619	0.428	0.606
	LSTM (raw RR)	0.426	0.516	0.341	0.567
	CNN (raw RR)	0.432	0.541	0.355	0.551
Autoencoder					
	SVM + AE (5 dim)	0.609	0.709	0.503	0.772
	SVM + PCA (10 comp)	0.719	0.825	0.680	0.763
	AE anomaly det.	0.454	0.348	0.301	0.921

5 Discussion

5.1 Interpretation of Key Findings

The central finding of this study (F1=0.778 under LOSO cross-validation) carries methodological weight precisely because of the evaluation protocol. Studies reporting comparable or higher accuracy, such as Hafiz et al. [8] (81.5%) and Butkeviciute et al. [2], used within-subject or random k-fold partitioning, which allows training on windows from the same participant later used for testing. This may inflate reported performance by conflating memorisation of individual physiological signatures with genuine generalisation. The present result, derived from complete participant separation at every fold, is a more conservative estimate of cross-subject generalisation, though real-world deployability would require validation beyond the controlled laboratory setting used here.

The dominance of *temp_change_from_baseline* across all models is the most consistently important feature in this dataset. It suggests that fatigue manifests as a change from an individual's own resting state rather than crossing a fixed physiological threshold. This may partly explain why HRV-only approaches underperform under LOSO as HRV features without a relative anchor are vulnerable to the substantial inter-individual differences in autonomic baseline documented by Shaffer and Ginsberg [26]. These findings suggest that fatigue detection systems may benefit from a calibration phase to establish participant-specific baselines before monitoring begins, though this inference is drawn from a small controlled study and requires further validation. It should be noted that *temp_change_from_baseline* is itself a form of implicit per-session calibration, computed relative to each participant's first window. The model therefore depends on a within-session resting reference point rather than being fully calibration-free, a dependency that may limit applicability in settings where a structured rest period cannot be guaranteed.

The second-ranked features, `sdhr` and `cv_rr`, reflect reduction in heart rate variability rather than elevation in mean heart rate. This is consistent with the vagal withdrawal mechanism described by Hilgarter et al. [10], in which sustained cognitive demand shifts sympathovagal balance toward sympathetic dominance, producing a more rigid and less reactive cardiac signal. The non-significance of `mean_hr` and `mean_rr` is notable and consistent with the baseline measurement design, in which participants performed reaction time and n-back tasks at M1, elevating heart rate to levels that overlap with the fatigued state. This confound is absent in studies using passive rest as baseline, which may partially explain performance differences in the literature.

It should also be acknowledged that the F1=0.778 result was obtained on a highly controlled dataset with a structured protocol, fixed cognitive tasks, and guaranteed resting baselines. It is unclear whether this performance would transfer to naturalistic settings where fatigue accumulates irregularly and resting periods cannot be enforced. The result should therefore be interpreted as an upper bound on what may be achievable under idealised conditions rather than an estimate of real-world performance.

5.2 The Role of Multimodal Fusion

The most direct comparison available in this study is the feature subset evaluation. HRV alone achieved F1=0.376, barely above chance, while temperature alone achieved F1=0.570. Combining the two modalities, the results reached F1=0.778. The super-additive improvement (ΔF1=+0.402 over HRV-only) is consistent with the possibility that cardiac autonomic features alone may be insufficient for cross-subject detection, though this requires replication in larger cohorts before broader conclusions can be drawn.

The physiological basis for this complementarity is interpretable. HRV features capture rapid, beat-to-beat autonomic dynamics that are discriminative within sessions but highly sensitive to inter-individual differences in autonomic reactivity, posture, and respiratory patterns [19], all of which vary across participants in LOSO. Skin temperature, by contrast, evolves slowly over minutes as sympathetically mediated peripheral vasoconstriction accumulates, making it a stable trend signal. Crucially, when expressed relative to each participant's own baseline, temperature is self-normalising in a way that raw HRV metrics are not. The combination therefore provides both moment-to-moment discriminative power from HRV and a participant-normalised fatigue trend from temperature. The self-normalising property of temperature, however, presupposes access to a stable resting period at session onset, which may not always be available in unconstrained monitoring scenarios.

An alternative explanation for temperature outperforming HRV under LOSO is that the FatigueSet protocol's structured baseline period systematically favours baseline-relative temperature features. In a dataset without a guaranteed resting period, this advantage may not hold, and HRV features may prove comparably or more discriminative. The relative contribution of each modality should therefore be interpreted with caution as potentially dataset-specific.

5.3 Traditional Machine Learning vs Deep Learning

The kernel SVM consistently outperformed all neural architectures, with raw RR interval models performing particularly poorly (CNN: F1=0.432, LSTM: F1=0.426). This is consistent with the known limitations of data-hungry neural architectures on small datasets, approximately 400 labelled windows across 12 participants provides insufficient data for end-to-end architectures to learn beat-to-beat temporal representations that generalise across individuals under LOSO. The CNN and LSTM models developed a systematic bias toward predicting fatigue, misclassifying over half of all baseline windows, suggesting they latched onto the lower variance of the fatigued RR signal as a heuristic that broke down for unseen participants.

The 22 hand-crafted HRV features encode decades of autonomic physiology research into compact discriminative statistics, performing a domain-informed dimensionality reduction that models trained from 400 samples may struggle to reliably replicate from raw signals without domain guidance. The window size experiments support this interpretation whereby tripling training samples by moving to 30-second windows did not enable raw RR models to close the gap with SVM, suggesting the bottleneck is the absence of physiological inductive bias rather than data volume. The MLP applied to the same features (F1=0.735) approached but did not exceed the SVM, consistent with the SVM's known sample efficiency advantage under small training fold sizes, attributable to its maximum-margin formulation rather than gradient-based optimisation. These findings suggest that for physiological fatigue datasets at this scale, investment in domain-specific feature engineering may outperform architectural complexity. It should also be noted that the MLP architecture used here was relatively shallow and fixed; more sophisticated neural architectures or regularisation strategies were not explored. Furthermore, the aggregate F1 score across 12 LOSO folds may mask high variance between folds. With only 12 participants, a single fold with an atypical participant can substantially shift the aggregate metric, meaning the reported F1=0.778 may not be a stable estimate of expected performance on new individuals.

5.4 Failure of Unsupervised Approaches

K-Means achieved 52.8% cluster alignment, marginally above chance. This is not a failure of implementation but reflects a potential mismatch between unsupervised assumptions and the phenomenology of mental fatigue. Clustering methods implicitly assume the target state is more distinctive or separable than baseline. Mental fatigue produces the opposite: it is a more uniform, lower-variance physiological signature. Without supervised context (knowledge of which windows preceded and followed cognitive exertion) this reduced variability is indistinguishable from a calm resting state. These results suggest that purely unsupervised fatigue detection may be unreliable, at least for HRV and temperature signals at this dataset scale, and ground-truth labelled data capturing both states may be practically necessary for the approach taken here. More sophisticated unsupervised approaches such as Gaussian Mixture Models or density-based clustering were not evaluated and may yield different conclusions; the failure observed here is specific to K-Means on this feature set and should not be generalised beyond it.

5.5 Limitations

Three limitations warrant acknowledgement. First, with 12 participants, inter-individual variability dominates the LOSO results as four participants showed F1 below 0.600, while three showed near-perfect classification. This spread makes it difficult to distinguish whether weaker participants represent genuinely atypical

fatigue physiology, inadequate feature generalisation, or insufficient data at their folds. A larger cohort would allow systematic investigation of which participant characteristics predict poor cross-subject generalisation, which is arguably more actionable than aggregate performance improvement. The perfect classification of P03 (F1=1.000) warrants particular caution. While it may reflect genuinely consistent fatigue physiology, it could alternatively indicate that this participant's physiological profile was unusually well represented in the training data, a possibility that cannot be ruled out at this sample size.

Second, the binary baseline-versus-fatigued framing, while experimentally clean, is a simplification. The three-class analysis found that only three features, `sdhr`, `cv_rr`, and `max_hr`, distinguished all three fatigue levels, suggesting that fine-grained fatigue staging remains an open problem with the current feature set. Real-world fatigue exists on a continuum and may more appropriately motivate regression formulations or ordinal classification, particularly as the boundary between mild and full fatigue was physiologically ambiguous in this dataset.

Third, the controlled laboratory setting imposes limits on external validity. Real-world mental fatigue accumulates through heterogeneous cognitive activities, varying environmental conditions, and concurrent physical states, none of which are captured here. The 180-second window required for stable frequency-domain HRV estimation also imposes a minimum latency that may be impractical for real-time alerting systems requiring rapid response. Additionally, the high variance in VAS mental fatigue scores at M3 (36.4±21.9) suggests that the cognitive dual task did not induce equivalent levels of fatigue across all participants, meaning some labelled fatigued windows may not reflect genuine mental fatigue, potentially introducing label noise that could affect both classification performance and the interpretation of feature importance results.

Finally, a methodological limitation of the statistical analysis is that Mann-Whitney U tests were applied to all 402 windows as independent observations, despite windows from the same participant being temporally correlated due to the 66.7% overlap and shared individual physiology. This violates the strict independence assumption of the test, inflating the effective sample size and making the reported p-values anticonservative. A more rigorous approach would aggregate features to participant level before testing, though this would reduce the sample to 12 observations per group. Additionally, 28 simultaneous tests were conducted without multiple comparisons correction, meaning approximately one false positive would be expected by chance at $p<0.05$. The reported p-values should therefore be interpreted as exploratory indicators of discriminative potential rather than confirmatory evidence, with effect sizes and the converging classification results providing stronger support for the findings.

5.6 Practical Implications

These findings carry specific implications for wearable fatigue detection design. The results are consistent with a multimodal sensing approach being beneficial for devices that combine cardiac monitoring with peripheral temperature, a combination already available on consumer wearables. The importance of participant-relative normalisation implies that a structured resting period at session onset is a practical requirement, given that the model's strongest feature depends on it, rather than an optional addition. Finally, the under-performance of deep learning at this scale suggests that domain-specific feature engineering with kernel methods may be preferable for datasets of comparable size, though this is an observation from a single study rather than a prescriptive guideline. As larger fatigue datasets become available, deep learning approaches should be revisited, but the present results suggest this is not yet the binding constraint.

6 Conclusions

This study investigated mental fatigue detection from HRV and wrist skin temperature features using the FatigueSet dataset, evaluating a complete pipeline from preprocessing through to deep learning and machine learning under Leave-One-Subject-Out cross-validation. The best configuration, an SVM with a reduced 13-feature set ($C = 5.0$, RBF kernel), achieved F1=0.778 and accuracy=0.863, suggesting that reliable cross-subject fatigue detection may be feasible from wearable physiological signals.

Four key findings emerged. First, `temp_change_from_baseline` was the single most discriminative feature across all models, suggesting that participant-relative peripheral temperature drift may be a stronger fatigue indicator than individual HRV measures in this dataset, a result that has direct implications for how deployed systems should handle inter-individual physiological variability. Second, multimodal fusion was essential as HRV alone achieved F1=0.376 versus F1=0.778 combined, with the super-additive improvement suggesting that cardiac and thermoregulatory signals capture complementary aspects of the fatigue response, though replication in larger cohorts is needed to confirm this. Third, traditional machine learning with hand-crafted features substantially outperformed all deep learning architectures at this dataset scale, with raw RR interval models failing to exceed F1=0.502; the bottleneck appears to be the absence of physiological inductive bias rather than data volume, as indicated by window size experiments that tripled training samples without closing the performance gap, though this conclusion is specific to the dataset scale used here. Fourth, unsupervised clustering failed to separate baseline from fatigued states (52.8% alignment), reflecting a potential incompatibility between clustering assumptions and the reduced-variability nature of mental fatigue, suggesting that labelled training data capturing both states may be a practical necessity for this detection task, at least with the signals and methods evaluated here.

Future work should prioritise larger participant cohorts to provide the statistical power needed for robust cross-subject generalisation conclusions and to revisit deep learning approaches under more favourable data conditions. Additionally, future studies should consider collecting data in naturalistic settings rather than relying solely on controlled laboratory datasets, to improve ecological validity. Multi-class and regression-based formulations of fatigue represent a natural extension, given that the three-class statistical analysis identified only three features capable of distinguishing all fatigue levels simultaneously. Personalised adaptive models that continuously update participant-specific baselines represent a promising direction for addressing the persistent inter-individual variability observed. Real-world deployment studies are ultimately required to validate whether laboratory-derived physiological signatures transfer to naturalistic cognitive workload settings.

Acknowledgments

I would like to thank my supervisor, Dr Fahim Kawsar, for the invaluable guidance and support throughout this study.

References

[1] Maarten A.S. Boksem and Mattie Tops. 2008. Mental fatigue: Costs and benefits. *Brain Research Reviews* 59, 1 (2008), 125–139. doi:10.1016/j.brainresrev.2008.07.001

[2] Eglė Butkevičiūtė, Aleksėjus Michalkovič, and Liepa Bikulčienė. 2022. ECG Signal Features Classification for the Mental Fatigue Recognition. *Mathematics* 10, 18 (2022). doi:10.3390/math10183395

[3] John A. Caldwell, J. Lynn Caldwell, and Regina M. Schmidt. 2008. Alertness management strategies for operational contexts. *Sleep Medicine Reviews* 12, 4 (2008), 257–273. doi:10.1016/j.smrv.2008.01.002

[4] Nitesh V. Chawla, Kevin W. Bowyer, Lawrence O. Hall, and W. Philip Kegelmeyer. 2002. SMOTE: synthetic minority over-sampling technique. *J. Artif. Int. Res.* 16, 1 (June 2002), 321–357.

[5] Galya Georgieva-Tsaneva, Krasimir Cheshmedzhiev, Yoan-Aleksandar Tsanev, and Miroslav Dechev. 2025. Physiological State Recognition via HRV and Fractal Analysis Using AI and Unsupervised Clustering. *Information* 16, 9 (2025). doi:10.3390/info16090718

[6] Martin Gjoreski, Mitja Luštrek, Matjaž Gams, and Hristijan Gjoreski. 2017. Monitoring stress with a wrist device using context. *Journal of Biomedical Informatics* 73 (2017), 159–170. doi:10.1016/j.jbi.2017.08.006

[7] Dalong Guo, Cong Wang, Yufei Qin, Lamei Shang, Aijing Gao, Baosen Tan, Yubin Zhou, and Guangyun Wang. 2025. Assessment of flight fatigue using heart rate variability and machine learning approaches. *Frontiers in Neuroscience* Volume 19 - 2025 (2025). doi:10.3389/fnins.2025.1621638

[8] Wynona Salsabila Hafiz, Maya Arlini Puspasari, Dewi Yunia Fitriani, Richard Joseph Hanowski, Danu Hadi Syaifullah, and Salsabila Annisa Arista. 2025. Developing a Fatigue Detection Model for Hospital Nurses Using HRV Measures and Machine Learning. *Safety* 11, 2 (2025). doi:10.3390/safety11020048

[9] Katherine A. Herborn, James L. Graves, Paul Jerem, Neil P. Evans, Ruedi Nager, Dominic J. McCafferty, and Dorothy E.F. McKeegan. 2015. Skin temperature reveals the intensity of acute stress. *Physiology Behavior* 152 (2015), 225–230. doi:10.1016/j.physbeh.2015.09.032

[10] Kathrin Hilgarter, Karin Schmid-Zalaudek, Regina csanady leitner, Manfred Mörtl, Andreas Roessler, and Helmut Lackner. 2021. Phasic heart rate variability and the association with cognitive performance: A cross-sectional study in a healthy population setting. *PLOS ONE* 16 (03 2021). doi:10.1371/journal.pone.0246968

[11] Niraj Hirachan, Anita Mathews, Julio Romero, and Raul Fernandez Rojas. 2022. Measuring Cognitive Workload Using Multimodal Sensors. In *2022 44th Annual International Conference of the IEEE Engineering in Medicine & Biology Society (EMBC)*. 4921–4924. doi:10.1109/EMBC48229.2022.9871308

[12] Sergey Ioffe and Christian Szegedy. 2015. Batch normalization: accelerating deep network training by reducing internal covariate shift. In *Proceedings of the 32nd International Conference on Machine Learning - Volume 37* (Lille, France) *(ICML'15)*. JMLR.org, 448–456.

[13] Manasa Kalanadhabhatta, Chulhong Min, Alessandro Montanari, and Fahim Kawsar. 2022. FatigueSet: A Multi-modal Dataset for Modeling Mental Fatigue and Fatigability. In *Pervasive Computing Technologies for Healthcare*, Hadas Lewy and Refael Barkan (Eds.). Springer International Publishing, Cham, 204–217.

[14] Diederik P. Kingma and Jimmy Ba. 2015. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations (ICLR)*. https://arxiv.org/abs/1412.6980

[15] K. Kunasegaran, A. M. H. Ismail, S. Ramasamy, J. V. Gnanou, B. A. Caszo, and P. L. Chen. 2023. Understanding mental fatigue and its detection: a comparative analysis of assessments and tools. *PeerJ* 11 (Aug. 2023), e15744. doi:10.7717/peerj.15744

[16] Marek Malik. 1996. Heart rate variability: Standards of measurement, physiological interpretation, and clinical use. *Circulation* 93 (03 1996), 1043–1065.

[17] Enrique Molina, Daniel Sanabria, Tzyy-Ping Jung, and Ángel Correa. 2019. Electroencephalographic and peripheral temperature dynamics during a prolonged psychomotor vigilance task. *Accident Analysis Prevention* 126 (2019), 198–208. doi:10.1016/j.aap.2017.10.014 10th International Conference on Managing Fatigue: Managing Fatigue to Improve Safety, Wellness, and Effectiveness"..

[18] Alessandro Montanari, A. Thangarajan, Khaldoon Al-Naimi, Andrea Ferlini, Yang Liu, A. Balaji, and F. Kawsar. 2024. OmniBuds: A Sensory Earable Platform for Advanced Bio-Sensing and On-Device Machine Learning. *ArXiv* abs/2410.04775 (2024). doi:10.48550/arxiv.2410.04775

[19] Markus Muehlhan, Michael Marxen, Julia Landsiedel, Hagen Malberg, and Sebastian Zaunseder. 2014. The effect of body posture on cognitive performance: a question of sleep quality. *Frontiers in Human Neuroscience* 8 (03 2014), 171. doi:10.3389/fnhum.2014.00171

[20] P. Zihisire Muke and Adrianna Kozierkiewicz. 2025. Machine Learning Techniques to Improve the Cognitive Workload Classification Using Multimodal Sensors' Data. *IEEE Access* 13 (2025), 173415–173443. doi:10.1109/ACCESS.2025.3616788

[21] Ashley N. Nielsen, Deanna M. Barch, Steven E. Petersen, Bradley L. Schlaggar, and Deanna J. Greene. 2020. Machine Learning With Neuroimaging: Evaluating Its Applications in Psychiatry. *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* 5, 8 (2020), 791–798. doi:10.1016/j.bpsc.2019.11.007 Understanding the Nature and Treatment of Psychopathology: Letting the Data Guide the Way.

[22] Sophie Maria Penfold, James Cunningham, Pauline Whelan, Martin G. McCabe, and John Ainsworth. 2025. Decreased Heart Rate Variability Is Associated with Increased Fatigue Across Different Medical Populations: A Systematic Review. *Pathophysiology* 32, 3 (2025). doi:10.3390/pathophysiology32030046

[23] Todd Pickering, Bradley Wright, and Clare MacMahon. 2024. Fatigued or bored? Investigating the effect of different types of mental fatigue on 3 km running performance. *Psychology of Sport and Exercise* 74 (2024), 102687. doi:10.1016/j.psychsport.2024.102687

[24] N. Romeijn, R. J. Raymann, E. Møst, et al. 2012. Sleep, vigilance, and thermosensitivity. *Pflügers Archiv* 463, 1 (2012), 169–176. doi:10.1007/s00424-011-1042-2

[25] Eike A. Schmidt, Michael Schrauf, Michael Simon, Martin Fritzsche, Axel Buchner, and Wilhelm E. Kincses. 2009. Drivers' misjudgement of vigilance state during prolonged monotonous daytime driving. *Accident Analysis Prevention* 41, 5 (2009), 1087–1093. doi:10.1016/j.aap.2009.06.007

[26] Fred Shaffer and J. P. Ginsberg. 2017. An Overview of Heart Rate Variability Metrics and Norms. *Frontiers in Public Health* Volume 5 - 2017 (2017). doi:10.3389/fpubh.2017.00258

[27] J. Shen, J. Barbera, and C. M. Shapiro. 2006. Distinguishing sleepiness and fatigue: focus on definition and measurement. *Sleep Medicine Reviews* 10, 1 (Feb. 2006), 63–76. doi:10.1016/j.smrv.2005.05.004

[28] Nitish Srivastava, Geoffrey Hinton, Alex Krizhevsky, Ilya Sutskever, and Ruslan Salakhutdinov. 2014. Dropout: a simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.* 15, 1 (Jan. 2014), 1929–1958.

[29] Joachim Taelman, S. Vandeput, A. Spaepen, and S. Van Huffel. 2009. Influence of Mental Stress on Heart Rate and Heart Rate Variability. In *4th European Conference of the International Federation for Medical and Biological Engineering*, Jos Vander Sloten, Pascal Verdonck, Marc Nyssen, and Jens Haueisen (Eds.). Springer Berlin Heidelberg, Berlin, Heidelberg, 1366–1369.

[30] Y. Tran, A. Craig, R. Craig, R. Chai, and H. Nguyen. 2020. The influence of mental fatigue on brain activity: Evidence from a systematic review with meta-analyses. *Psychophysiology* 57, 5 (May 2020), e13554. doi:10.1111/psyp.13554

[31] World Health Organization. 2019. Burn-out an "occupational phenomenon": International Classification of Diseases. https://www.who.int/news/item/28-05-2019-burn-out-an-occupational-phenomenon-international-classification-of-diseases Accessed: 2025.

A Label Sequence Validation

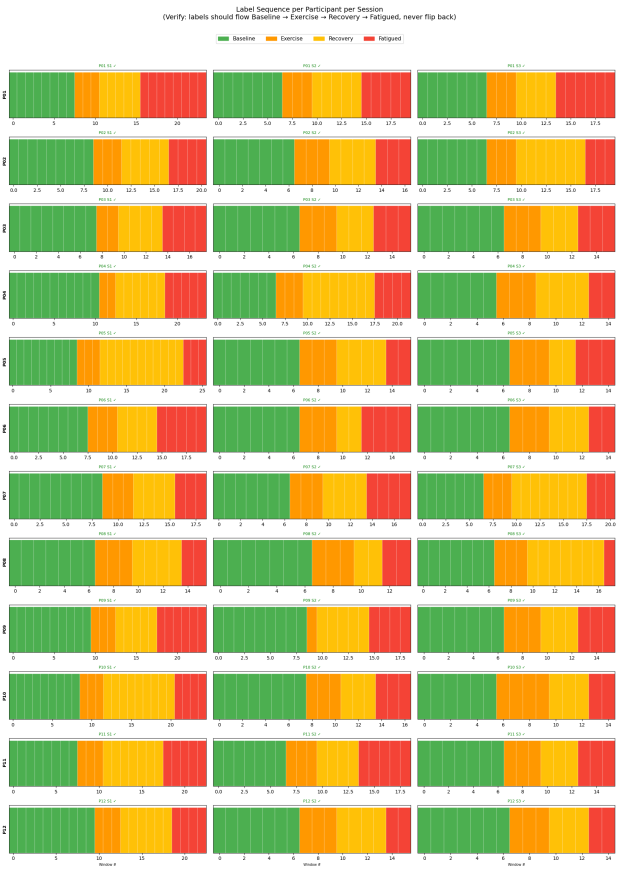


Figure 16: Label sequence per participant per session across all 36 sessions. Colours indicate experimental phase: baseline (green), exercise (orange), recovery (yellow), fatigued (red). All sessions follow the expected chronological progression with no flip-flopping detected.

B PCA Explained Variance

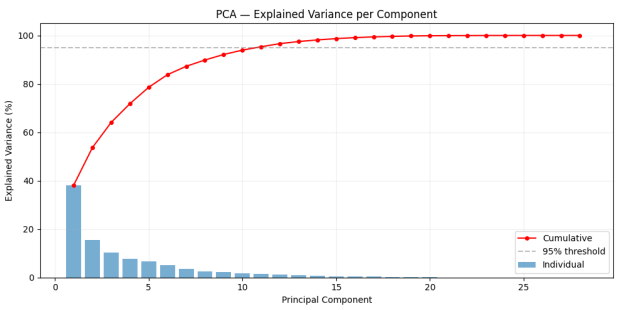


Figure 17: PCA explained variance per component for the 28-feature set. Eleven components are required to capture 95% of total variance, suggesting the feature space is genuinely high-dimensional with no dominant low-dimensional structure.

C Per-Model Feature Ablation Plots

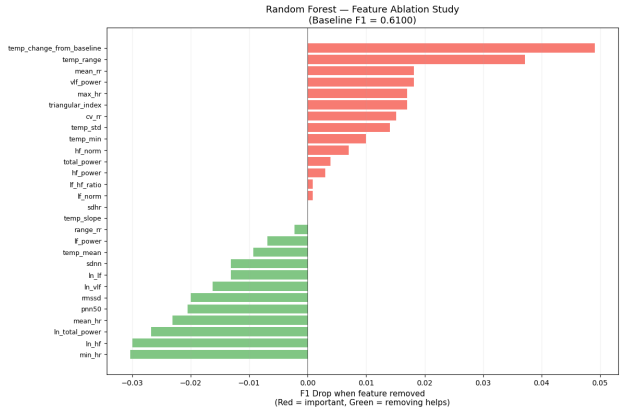


Figure 18: Random Forest feature ablation (baseline F1=0.610).

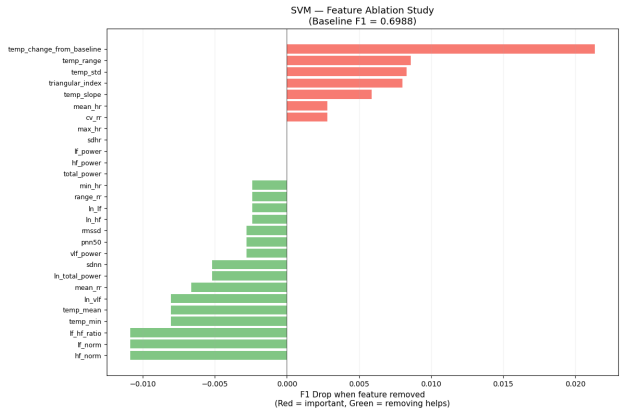


Figure 19: SVM feature ablation (baseline F1=0.699).

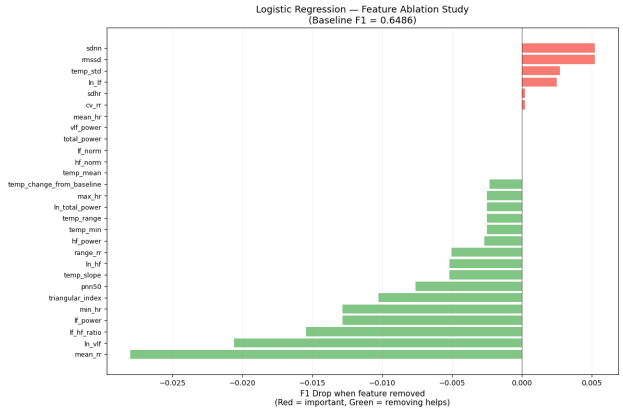


Figure 20: Logistic Regression feature ablation (baseline F1=0.649).

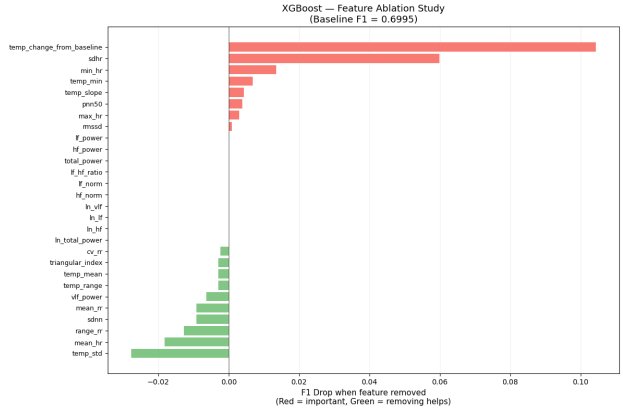


Figure 21: XGBoost feature ablation (baseline F1=0.700).

D Baseline Model Confusion Matrix

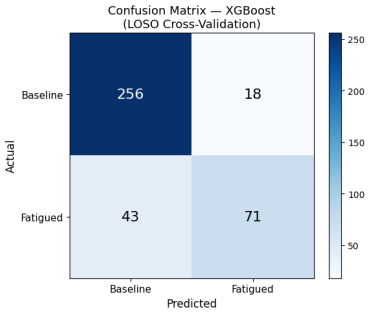


Figure 22: Confusion matrix for XGBoost (best model by accuracy) on the full 28-feature set under LOSO cross-validation. Of 127 fatigued windows, 71 were correctly classified (recall=0.623), while 256 of 275 baseline windows were correctly identified (specificity=0.931).

E Confusion Matrix for Best Model

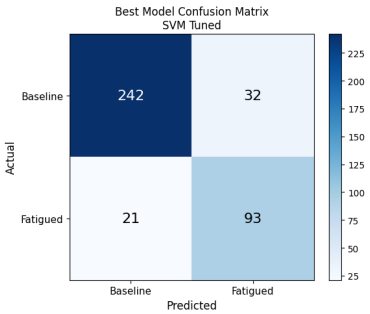


Figure 23: Confusion matrix for the best model configuration (SVM, reduced 13 features, $C = 5.0$, RBF kernel) under LOSO cross-validation. Of 127 fatigued windows, 103 were correctly classified (recall=0.816), while 257 of 275 baseline windows were correctly identified (specificity=0.935).

F Per-Session Prediction Sequences

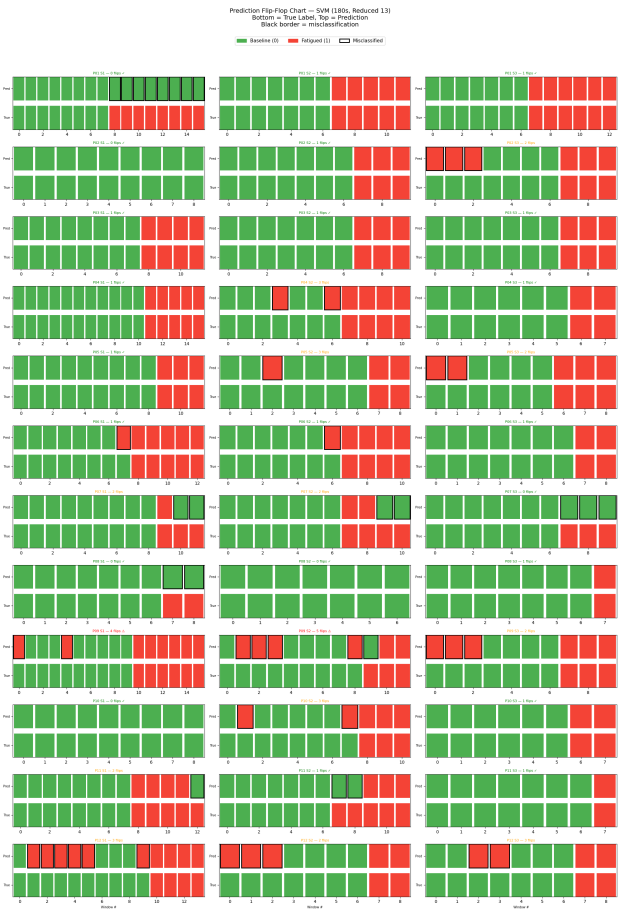


Figure 24: Prediction sequences for the best model (SVM, reduced 13 features, $C = 5.0$, RBF) across all 36 sessions. Bottom row shows true labels, top row shows predictions. Black borders indicate misclassified windows. Sessions with orange titles indicate flip-flop predictions. Most errors occur at the baseline-to-fatigued transition boundary.

G Deep Learning Confusion Matrices

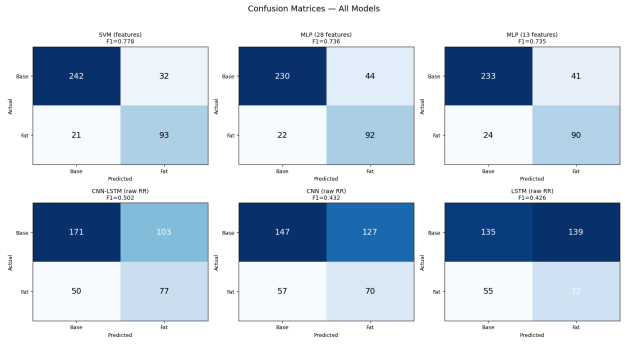


Figure 25: Confusion matrices for all six models under LOSO cross-validation. Feature-based models (SVM, MLP) correctly identify the majority of fatigued windows while keeping false positives low. Raw RR models exhibit high false positive rates, misclassifying large proportions of baseline windows as fatigued.

H Labelling Strategy Confusion Matrices

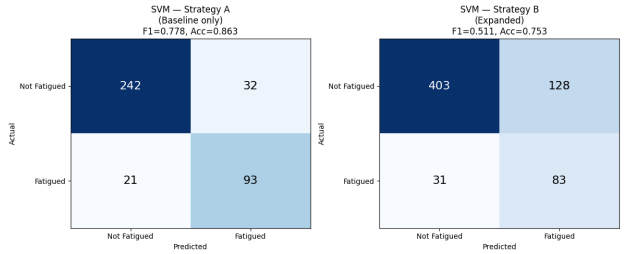


Figure 26: Confusion matrices for SVM under Strategy A (left) and Strategy B (right). Strategy B increases both false positives (128 vs 32) and false negatives (31 vs 21), simultaneously degrading precision and recall. The high accuracy of Strategy B (0.753) despite low F1 (0.511) reflects the model biasing towards the majority non-fatigued class.

I Labelling Strategy Comparison

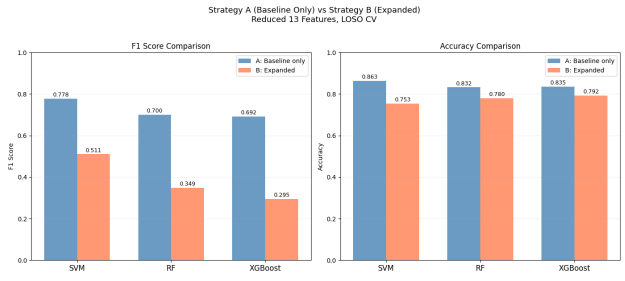


Figure 27: F1 score and accuracy comparison between Strategy A (baseline only as negative class) and Strategy B (baseline, exercise, and recovery as negative class). Strategy A consistently outperforms Strategy B across all three models, validating the decision to exclude exercise and recovery windows from training.

J Random Forest Gini Feature Importance

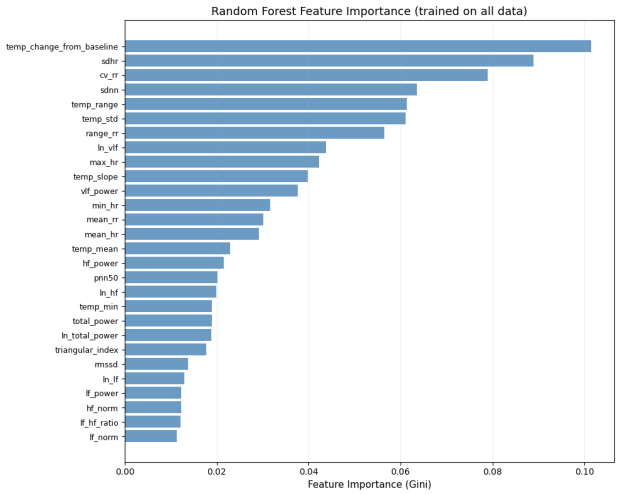


Figure 28: Random Forest Gini feature importance trained on all data. Results corroborate the ablation study, with temperature-derived features and HRV variability measures occupying the top rankings. *temp_change_from_baseline* ranks highest, consistent with its dominant F1 drop in the ablation analysis across all four models.

K Baseline vs Improved Model Comparison

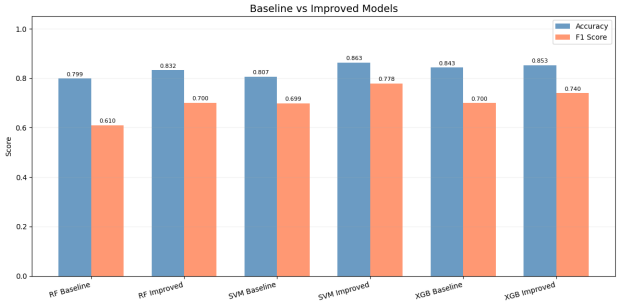


Figure 29: Comparison of baseline and improved model performance. SVM shows the largest improvement (ΔF1= +0.079) following hyperparameter tuning and feature reduction.

L Feature Subset Comparison

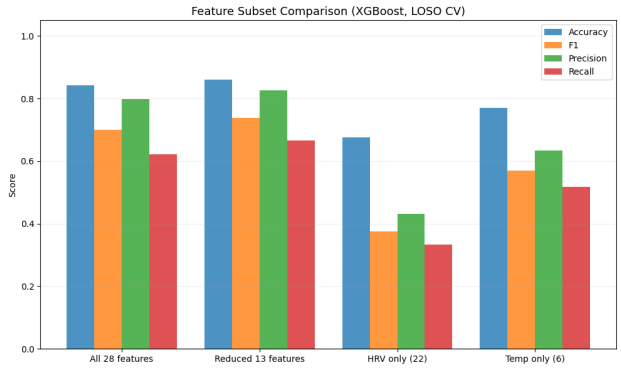


Figure 30: Feature subset comparison using XGBoost under LOSO cross-validation. The reduced 13-feature set outperforms the full 28-feature set, while HRV-only features perform substantially worse than temperature-only, supporting the importance of multimodal fusion.

M Raw RR Interval Analysis

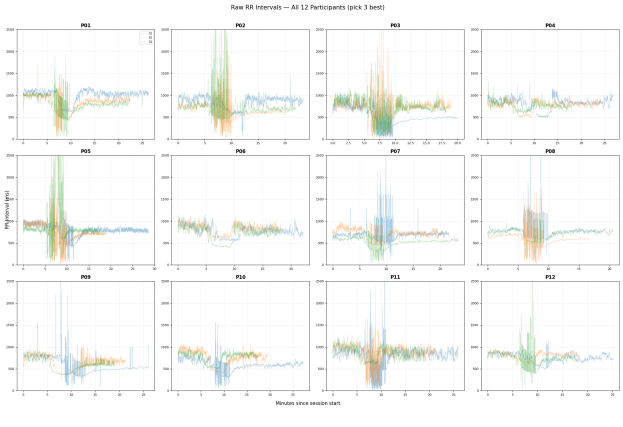


Figure 31: Raw RR interval distributions across all sessions for each of the 12 participants.

N HRV Feature Extraction and Raw Plot Comparison for Three Participants

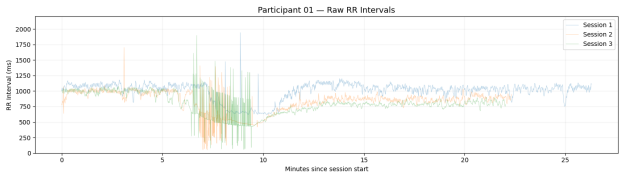


Figure 32: Raw RR interval time series for Participant 1.

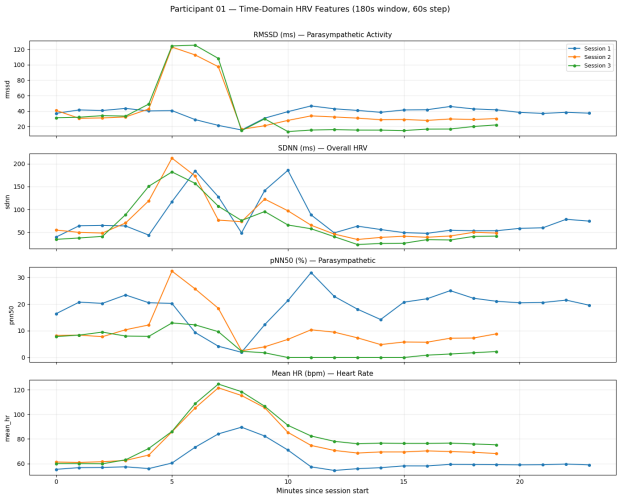


Figure 33: Time-domain HRV features extracted for Participant 1.



Figure 34: Frequency-domain HRV features extracted for Participant 1.

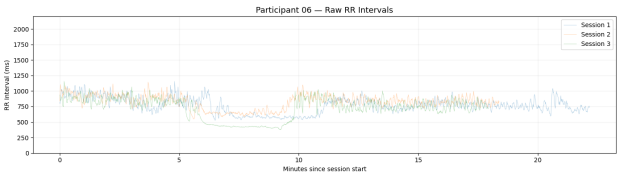


Figure 35: Raw RR interval time series for Participant 6.

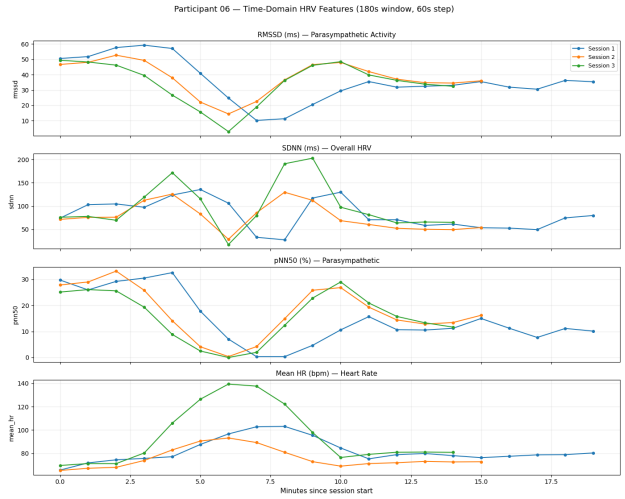


Figure 36: Time-domain HRV features extracted for Participant 6.



Figure 37: Frequency-domain HRV features extracted for Participant 6.

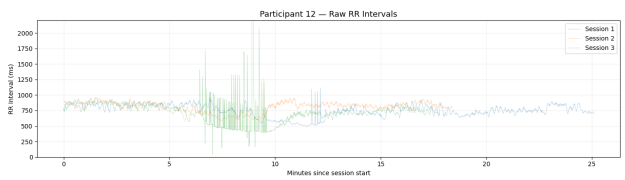


Figure 38: Raw RR interval time series for Participant 12.

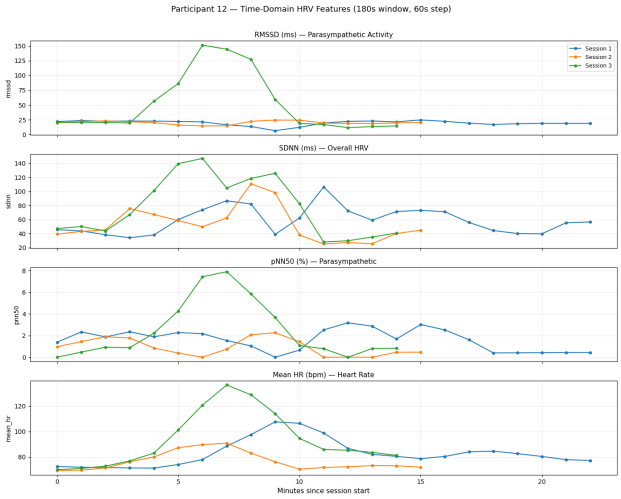


Figure 39: Time-domain HRV features extracted for Participant 12.



Figure 40: Frequency-domain HRV features extracted for Participant 12.

O Three-Class Statistical Analysis

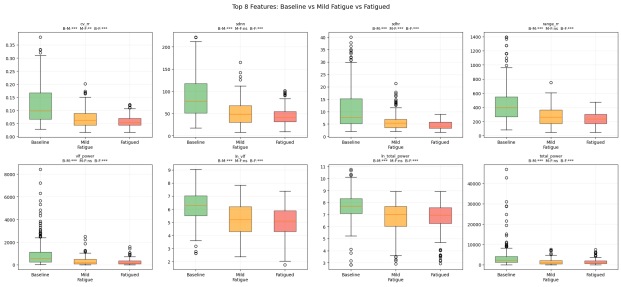


Figure 41: Box plots for the top 8 features across three classes: baseline, mild fatigue, and fatigued.

P Temperature Feature Extraction for Three Participants

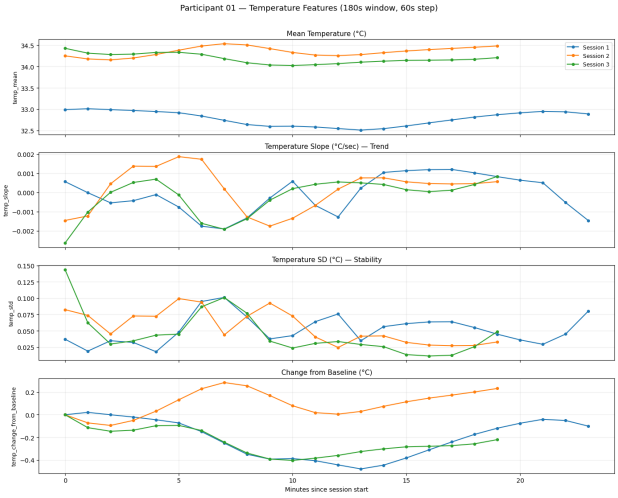


Figure 42: Skin temperature features extracted for Participant 1.



Figure 44: Skin temperature features extracted for Participant 12.

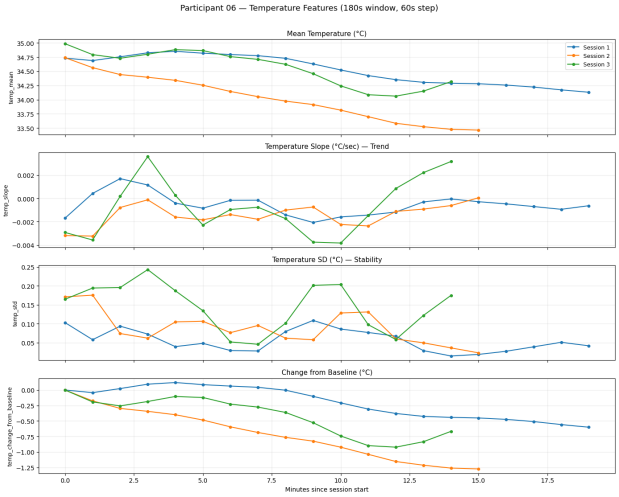


Figure 43: Skin temperature features extracted for Participant 6.